Neural Computation -- 2000

140 article(s), all issues

1. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2000.12.issue-4

No abstract available.

---

2. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2000.12.issue-6

No abstract available.

---

3. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2000.12.issue-5

No abstract available.

---

4. (untitled)

Knight, Bruce W.

https://doi.org/10.1162/neco.2000.12.issue-3

The use of a population dynamics approach promises efficient simulation of large assemblages of neurons. Depending on the issues addressed and the degree of realism incorporated in the simulated neurons, a wide range of different population dynamics formulations can be appropriate. Here we present a common mathematical structure that these various formulations share and that implies dynamical behaviors that they have in common. This underlying structure serves as a guide toward efficient means of simulation. As an example, we derive the general population firing-rate frequency-response and show how it may be used effectively to address a broad range of interacting-population response and stability problems. A few specific cases will be worked out. A summary of this work appears at the end, before the appendix.

---

5. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2000.12.issue-1

No abstract available.

---

6. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2000.12.issue-11

No abstract available.

---

7. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2000.12.issue-12

No abstract available.

---

8. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2000.12.issue-10

No abstract available.

---

9. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2000.12.issue-7

No abstract available.

---

10. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2000.12.issue-9

No abstract available.

---

11. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2000.12.issue-2

No abstract available.

---

12. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2000.12.issue-8

No abstract available.

---

13. Correctness of Local Probability Propagation in Graphical Models with Loops

Yair Weiss

https://doi.org/10.1162/089976600300015880 | pp. 1-41

Graphical models, such as Bayesian networks and Markov networks, represent joint distributions over a set of variables by means of a graph. When the graph is singly connected, local propagation rules of the sort proposed by Pearl (1988) are guaranteed to converge to the correct posterior probabilities. Recently a number of researchers have empirically demonstrated good performance of these same local propagation schemes on graphs with loops, but a theoretical understanding of this performance has yet to be achieved. For graphical models with a single loop, we derive an analytical relationship between the probabilities computed using local propagation and the correct marginals. Using this relationship we show a category of graphical models with loops for which local propagation gives rise to provably optimal maximum a posteriori assignments (although the computed marginals will be incorrect). We also show how nodes can use local information in the messages they receive in order to correct their computed marginals. We discuss how these results can be extended to graphical models with multiple loops and show simulation results suggesting that some properties of propagation on single-loop graphs may hold for a larger class of graphs. Specifically we discuss the implication of our results for understanding a class of recently proposed error-correcting codes known as turbo codes.

---

14. Population Dynamics of Spiking Neurons: Fast Transients, Asynchronous States, and Locking

Wulfram Gerstner

https://doi.org/10.1162/089976600300015899 | pp. 43-89

An integral equation describing the time evolution of the population activity in a homogeneous pool of spiking neurons of the integrate-and-fire type is discussed. It is analytically shown that transients from a state of incoherent firing can be immediate. The stability of incoherent firing is analyzed in terms of the noise level and transmission delay, and a bifurcation diagram is derived. The response of a population of noisy integrate-and-fire neurons to an input current of small amplitude is calculated and characterized by a linear filter L. The stability of perfectly synchronized"locked"solutions is analyzed.

---

15. Dynamics of Strongly Coupled Spiking Neurons

Paul C. Bressloff, Stephen Coombes

https://doi.org/10.1162/089976600300015907 | pp. 91-129

We present a dynamical theory of integrate-and-fire neurons with strong synaptic coupling. We show how phase-locked states that are stable in the weak coupling regime can destabilize as the coupling is increased, leading to states characterized by spatiotemporal variations in the interspike intervals (ISIs). The dynamics is compared with that of a corresponding network of analog neurons in which the outputs of the neurons are

taken to be mean firing rates. A fundamental result is that for slow interactions, there is good agreement between the two models (on an appropriately defined timescale). Various examples of desynchronization in the strong coupling regime are presented. First, a globally coupled network of identical neurons with strong inhibitory coupling is shown to exhibit oscillator death in which some of the neurons suppress the activity of others. However, the stability of the synchronous state persists for very large networks and fast synapses. Second, an asymmetric network with a mixture of excitation and inhibition is shown to exhibit periodic bursting patterns. Finally, a one-dimensional network of neurons with long-range interactions is shown to desynchronize to a state with a spatially periodic pattern of mean firing rates across the network. This is modulated by deterministic fluctuations of the instantaneous firing rate whose size is an increasing function of the speed of synaptic response.

---

16. On Connectedness: A Solution Based on Oscillatory Correlation

DeLiang L. Wang

https://doi.org/10.1162/089976600300015916 | pp. 131-139

A long-standing problem in Neural Comp has been the problem of connectedness, first identified by Minsky and Papert (1969). This problem served as the cornerstone for them to establish analytically that perceptrons are fundamentally limited in computing geometrical (topological) properties. A solution to this problem is offered by a different class of neural networks: oscillator networks. To solve the problem, the representation of oscillatory correlation is employed, whereby one pattern is represented as a synchronized block of oscillators and different patterns are represented by distinct blocks that desynchronize from each other. Oscillatory correlation emerges from LEGION (locally excitatory globally inhibitory oscillator network), whose architecture consists of local excitation and global inhibition among neural oscillators. It is further shown that these oscillator networks exhibit sensitivity to topological structure, which may lay a neurocomputational foundation for explaining the psychophysical phenomenon of topological perception.

---

17. Practical Identifiability of Finite Mixtures of Multivariate Bernoulli Distributions

Miguel Á. Carreira-Perpiñán, Steve Renals

https://doi.org/10.1162/089976600300015925 | pp. 141-152

The class of finite mixtures of multivariate Bernoulli distributions is known to be nonidentifiable; that is, different values of the mixture parameters can correspond to exactly the same probability distribution. In principle, this would mean that sample estimates using this model would give rise to different interpretations. We give empirical support to the fact that estimation of this class of mixtures can still produce meaningful results in practice, thus lessening the importance of the identifiability problem. We also show that the expectation-maximization algorithm is guaranteed to converge to a proper maximum likelihood estimate, owing to a property of the log-likelihood surface. Experiments with synthetic data sets show that an original generating distribution can be estimated from a sample. Experiments with an electropalatography data set show important structure in the data.

---

18. The Effects of Pair-wise and Higher-order Correlations on the Firing Rate of a Postsynaptic Neuron

Sander M. Bohté, Henk Spekreijse, Pieter R. Roelfsema

https://doi.org/10.1162/089976600300015934 | pp. 153-179

Coincident firing of neurons projecting to a common target cell is likely to raise the probability of firing of this postsynaptic cell. Therefore, synchronized firing constitutes a significant event for postsynaptic neurons and is likely to play a role in neuronal information processing. Physiological data on synchronized firing in cortical networks are based primarily on paired recordings and cross-correlation analysis. However, pair-wise correlations among all inputs onto a postsynaptic neuron do not uniquely determine the distribution of simultaneous postsynaptic events. We develop a framework in order to calculate the amount of synchronous firing that, based on maximum entropy, should exist in a homogeneous neural network in which the neurons have known pair-wise correlations and higher-order structure is absent. According to the distribution of maximal entropy, synchronous events in which a large proportion of the neurons participates should exist even in the case of weak pair-wise correlations. Network simulations also exhibit these highly synchronous events in the case of weak pair-wise correlations. If such a group of neurons provides input to a common postsynaptic target, these network bursts may enhance the impact of this input, especially in the case of a high postsynaptic threshold. The proportion of neurons participating in synchronous bursts can be approximated by our method under restricted conditions. When these conditions are not fulfilled, the spike trains have less than maximal entropy, which is indicative of the presence of higher-order structure. In this situation, the degree of synchronicity cannot be derived from the pair-wise correlations.

---

19. Effects of Spike Timing on Winner-Take-All Competition in Model Cortical Circuits

Erik D. Lumer

https://doi.org/10.1162/089976600300015943 | pp. 181-194

Synaptic interactions in cortical circuits involve strong recurrent excitation between nearby neurons and lateral inhibition that is more widely spread. This architecture is commonly thought to promote a winner-take-all competition, in which a small fraction of neuronal responses is selected for further processing. Here I report that such a competition is remarkably sensitive to the timing of neuronal action potentials. This is shown using simulations of model neurons and synaptic connections representing a patch of cortical tissue. In the simulations, uncorrelated discharge among neuronal units results in patterns of response dominance and suppression, that is, in a winner-take-all competition. Synchronization of firing, however, prevents such competition. These results demonstrate a novel property of recurrent cortical-like circuits, suggesting that the temporal patterning of cortical activity may play an important part in selection among stimuli competing for the control of attention and motor action.

---

20. Model Dependence in Quantification of Spike Interdependence by Joint Peri-Stimulus Time Histogram

Hiroyuki Ito, S. Tsuji

https://doi.org/10.1162/089976600300015952 | pp. 195-217

Multineuronal recordings have enabled us to examine context-dependent changes in the relationship between the activities of multiple cells. The joint peri-stimulus time histogram (JPSTH) is a much-used method for investigating the dynamics of the interdependence of spike events between pairs of cells. Its results are often taken as an estimate of interaction strength between cells, independent of modulations in the cells' firing rates. We evaluate the adequacy of this estimate by examining the mathematical structure of how the JPSTH quantifies an interaction strength after excluding the contribution of firing rates. We introduce a simple probabilistic model of interacting point processes to generate simulated spike data and show that the normalized JPSTH incorrectly infers the temporal structure of variations in the interaction parameter strength. This occurs because, in our model, the correct normalization of firing-rate contributions is different than that used in Aertsen, Gerstein, Habib, and Palm's (1989) effective connectivity model. This demonstrates that firing-rate modulations cannot be corrected for in a model-independent manner, and therefore the effective connectivity does not represent a universal characteristic that is independent of modulation of the firing rates. Aertsen et al.'s (1989) effective connectivity may still be used in the analysis of experimental data, provided we are aware that this is simply one of many ways of describing the structure of interdependence. We also discuss some measure-independent characteristics of the structure of interdependence.

---

21. Reinforcement Learning in Continuous Time and Space

Kenji Doya

https://doi.org/10.1162/089976600300015961 | pp. 219-245

This article presents a reinforcement learning framework for continuous-time dynamical systems without a priori discretization of time, state, and action. Based on the Hamilton-Jacobi-Bellman (HJB) equation for infinite-horizon, discounted reward problems, we derive algorithms for estimating value functions and improving policies with the use of function approximators. The process of value function estimation is formulated as the minimization of a continuous-time form of the temporal difference (TD) error. Update methods based on backward Euler approximation and exponential eligibility traces are derived, and their correspondences with the conventional residual gradient, TD(0), and TD(lambda) algorithms are shown. For policy improvement, two methods-a continuous actor-critic method and a value-gradient-based greedy policy-are formulated. As a special case of the latter, a nonlinear feedback control law using the value gradient and the model of the input gain is derived. The advantage updating, a model-free algorithm derived previously, is also formulated in the HJB-based framework. The performance of the proposed algorithms is first tested in a nonlinear control task of swinging a pendulum up with limited torque. It is shown in the simulations that (1) the task is accomplished by the continuous actor-critic method in a number of trials several times fewer than by the conventional discrete actor-critic method; (2) among the continuous policy update methods, the value-gradient-based policy with a known or learned dynamic model performs several times better than the actor-critic method; and (3) a value function update using exponential eligibility traces is more efficient and stable than that based on Euler approximation. The algorithms are then tested in a higher-dimensional task: cart-pole swing-up. This task is accomplished in several hundred trials using the value-gradient-based policy with a learned dynamic model.

---

22. Minimizing Binding Errors Using Learned Conjunctive Features

Bartlett W. Mel, József Fiser
https://doi.org/10.1162/089976600300015772 | pp. 247-278

We have studied some of the design trade-offs governing visual representations based on spatially invariant conjunctive feature detectors, with an emphasis on the susceptibility of such systems to false-positive recognition errors-Malsburg's classical binding problem. We begin by deriving an analytical model that makes explicit how recognition performance is affected by the number of objects that must be distinguished, the number of features included in the representation, the complexity of individual objects, and the clutter load, that is, the amount of visual material in the field of view in which multiple objects must be simultaneously recognized, independent of pose, and without explicit segmentation. Using the domain of text to model object recognition in cluttered scenes, we show that with corrections for the nonuniform probability and nonindependence of text features, the analytical model achieves good fits to measured recognition rates in simulations involving a wide range of clutter loads, word size, and feature counts. We then introduce a greedy algorithm for feature learning, derived from the analytical model, which grows a representation by choosing those conjunctive features that are most likely to distinguish objects from the cluttered backgrounds in which they are embedded. We show that the representations produced by this algorithm are compact, decorrelated, and heavily weighted toward features of low conjunctive order. Our results provide a more quantitative basis for understanding when spatially invariant conjunctive features can support unambiguous perception in multiobject scenes, and lead to several insights regarding the properties of visual representations optimized for specific recognition tasks.

---

23. Relationship Between Phase and Energy Methods for Disparity Computation

Ning Qian, Samuel Mikaelian
https://doi.org/10.1162/089976600300015781 | pp. 279-292

The phase and energy methods for computing binocular disparity maps from stereograms are motivated differently, have different physiological relevances, and involve different computational steps. Nevertheless, we demonstrate that at the final stages where disparity values are made explicit, the simplest versions of the two methods are exactly equivalent. The equivalence also holds when the quadrature-pair construction in the energy method is replaced with a more physiologically plausible phase-averaging step. The equivalence fails, however, when the phase-difference receptive field model is replaced by the position-shift model. Additionally, intermediate results from the two methods are always quite distinct. In particular, the energy method generates a distributed disparity representation similar to that found in the visual cortex, while the phase method does not. Finally, more elaborate versions of the two methods are in general not equivalent. We also briefly compare these two methods with some other stereo models in the literature.

---

24. N-tuple Network, CART, and Bagging

Aleksander Ko■cz
https://doi.org/10.1162/089976600300015790 | pp. 293-304

Similarities between bootstrap aggregation (bagging) and N-tuple sampling are explored to propose a retina-free data-driven version of the N-tuple network, whose close analogies to aggregated regression

trees, such as classification and regression trees (CART), lead to further architectural enhancements. Performance of the proposed algorithms is compared with the traditional versions of the N-tuple and CART networks on a number of regression problems. The architecture significantly outperforms conventional N-tuple networks while leading to more compact solutions and avoiding certain implementational pitfalls of the latter.

---

25. Improving the Practice of Classifier Performance Assessment

Niall M. Adams, David J. Hand

https://doi.org/10.1162/089976600300015808 | pp. 305-311

In this note we use examples from the literature to illustrate some poor practices in assessing the performance of supervised classification rules, and we suggest guidelines for better methodology. We also describe a new assessment criterion that is suitable for the needs of many practical problems.

---

26. Do Simple Cells in Primary Visual Cortex Form a Tight Frame?

Emilio Salinas, L. F. Abbott

https://doi.org/10.1162/089976600300015817 | pp. 313-335

Sets of neuronal tuning curves, which describe the responses of neurons as functions of a stimulus, can serve as a basis for approximating other functions of stimulus parameters. In a function-approximating network, synaptic weights determined by a correlation-based Hebbian rule are closely related to the coefficients that result when a function is expanded in an orthogonal basis. Although neuronal tuning curves typically are not orthogonal functions, the relationship between function approximation and correlation-based synaptic weights can be retained if the tuning curves satisfy the conditions of a tight frame. We examine whether the spatial receptive fields of simple cells in cat and monkey primary visual cortex (V1) form a tight frame, allowing them to serve as a basis for constructing more complicated extrastriate receptive fields using correlation-based synaptic weights. Our calculations show that the set of V1 simple cell receptive fields is not tight enough to account for the acuity observed psychophysically.

---

27. Learning Overcomplete Representations

Michael S. Lewicki, Terrence J. Sejnowski

https://doi.org/10.1162/089976600300015826 | pp. 337-365

In an overcomplete basis, the number of basis vectors is greater than the dimensionality of the input, and the representation of an input is not a unique combination of basis vectors. Overcomplete representations have been advocated because they have greater robustness in the presence of noise, can be sparser, and can have greater flexibility in matching structure in the data. Overcomplete codes have also been proposed as a model of some of the response properties of neurons in primary visual cortex. Previous work has focused on finding the best representation of a signal using a fixed overcomplete basis (or dictionary). We present an algorithm for learning an overcomplete basis by viewing it as probabilistic model of the observed data. We show that overcomplete bases can yield a better approximation of the underlying statistical distribution of the data and can thus lead to greater coding efficiency. This can be viewed as a

generalization of the technique of independent component analysis and provides a method for Bayesian reconstruction of signals in the presence of noise and for blind source separation when there are more sources than mixtures.

---

28. Noise in Integrate-and-Fire Neurons: From Stochastic Input to Escape Rates

Hans Ekkehard Pleßer, Wulfram Gerstner

https://doi.org/10.1162/089976600300015835 | pp. 367-384

We analyze the effect of noise in integrate-and-fire neurons driven by time-dependent input and compare the diffusion approximation for the membrane potential to escape noise. It is shown that for time-dependent subthreshold input, diffusive noise can be replaced by escape noise with a hazard function that has a gaussian dependence on the distance between the (noise-free) membrane voltage and threshold. The approximation is improved if we add to the hazard function a probability current proportional to the derivative of the voltage. Stochastic resonance in response to periodic input occurs in both noise models and exhibits similar characteristics.

---

29. Modeling Synaptic Plasticity in Conjunction with the Timing of Pre- and Postsynaptic Action Potentials

Werner M. Kistler, J. Leo van Hemmen

https://doi.org/10.1162/089976600300015844 | pp. 385-405

We present a spiking neuron model that allows for an analytic calculation of the correlations between pre- and postsynaptic spikes. The neuron model is a generalization of the integrate-and-fire model and equipped with a probabilistic spike-triggering mechanism. We show that under certain biologically plausible conditions, pre- and postsynaptic spike trains can be described simultaneously as an inhomogeneous Poisson process. Inspired by experimental findings, we develop a model for synaptic long-term plasticity that relies on the relative timing of pre- and post-synaptic action potentials. Being given an input statistics, we compute the stationary synaptic weights that result from the temporal correlations between the pre- and postsynaptic spikes. By means of both analytic calculations and computer simulations, we show that such a mechanism of synaptic plasticity is able to strengthen those input synapses that convey precisely timed spikes at the expense of synapses that deliver spikes with a broad temporal distribution. This may be of vital importance for any kind of information processing based on spiking neurons and temporal coding.

---

30. On-line EM Algorithm for the Normalized Gaussian Network

Masa-aki Sato, Shin Ishii

https://doi.org/10.1162/089976600300015853 | pp. 407-432

A normalized gaussian network (NGnet) (Moody & Darken, 1989) is a network of local linear regression units. The model softly partitions the input space by normalized gaussian functions, and each local unit linearly approximates the output within the partition. In this article, we propose a new on-line EMalgorithm for the NGnet, which is derived from the batch EMalgorithm (Xu, Jordan, &Hinton 1995), by introducing a

discount factor. We show that the on-line EM algorithm is equivalent to the batch EM algorithm if a specific scheduling of the discount factor is employed. In addition, we show that the on-line EM algorithm can be considered as a stochastic approximation method to find the maximum likelihood estimator. A new regularization method is proposed in order to deal with a singular input distribution. In order to manage dynamic environments, where the input-output distribution of data changes over time, unit manipulation mechanisms such as unit production, unit deletion, and unit division are also introduced based on probabilistic interpretation. Experimental results show that our approach is suitable for function approximation problems in dynamic environments. We also apply our on-line EM algorithm to robot dynamics problems and compare our algorithm with the mixtures-of-experts family.

---

31. A General Probability Estimation Approach for Neural Computation

Maxim Khaikine, Klaus Holthausen

https://doi.org/10.1162/089976600300015862 | pp. 433-450

We describe an analytical framework for the adaptations of neural systems that adapt its internal structure on the basis of subjective probabilities constructed by computation of randomly received input signals. A principled approach is provided with the key property that it defines a probability density model that allows studying the convergence of the adaptation process. In particular, the derived algorithm can be applied for approximation problems such as the estimation of probability densities or the recognition of regression functions. These approximation algorithms can be easily extended to higher-dimensional cases. Certain neural network models can be derived from our approach (e.g., topological feature maps and associative networks).

---

32. On the Synthesis of Brain-State-in-a-Box Neural Models with Application to Associative Memory

Sevrani Fation, Kennichi Abe

https://doi.org/10.1162/089976600300015871 | pp. 451-472

In this article we present techniques for designing associative memories to be implemented by a class of synchronous discrete-time neural networks based on a generalization of the brain-state-in-a-box neural model. First, we address the local qualitative properties and global qualitative aspects of the class of neural networks considered. Our approach to the stability analysis of the equilibrium points of the network gives insight into the extent of the domain of attraction for the patterns to be stored as asymptotically stable equilibrium points and is useful in the analysis of the retrieval performance of the network and also for design purposes. By making use of the analysis results as constraints, the design for associative memory is performed by solving a constraint optimization problem whereby each of the stored patterns is guaranteed a substantial domain of attraction. The performance of the designed network is illustrated by means of three specific examples.

---

33. Dynamics of Encoding in Neuron Populations: Some General Mathematical Features

Bruce W. Knight

https://doi.org/10.1162/089976600300015673 | pp. 473-518

The use of a population dynamics approach promises efficient simulation of large assemblages of neurons. Depending on the issues addressed and the degree of realism incorporated in the simulated neurons, a wide range of different population dynamics formulations can be appropriate. Here we present a common mathematical structure that these various formulations share and that implies dynamical behaviors that they have in common. This underlying structure serves as a guide toward efficient means of simulation. As an example, we derive the general population firing-rate frequency-response and show how it may be used effectively to address a broad range of interacting-population response and stability problems. A few specific cases will be worked out. A summary of this work appears at the end, before the appendix.

---

34. Local and Global Gating of Synaptic Plasticity

Manuel Sánchez-Montañés, Paul F. M. J. Verschure, Peter König

https://doi.org/10.1162/089976600300015682 | pp. 519-529

Mechanisms influencing learning in neural networks are usually investigated on either a local or a global scale. The former relates to synaptic processes, the latter to unspecific modulatory systems. Here we study the interaction of a local learning rule that evaluates coincidences of pre- and postsynaptic action potentials and a global modulatory mechanism, such as the action of the basal forebrain onto cortical neurons. The simulations demonstrate that the interaction of these mechanisms leads to a learning rule supporting fast learning rates, stability, and flexibility. Furthermore, the simulations generate two experimentally testable predictions on the dependence of backpropagating action potential on basal forebrain activity and the relative timing of the activity of inhibitory and excitatory neurons in the neocortex.

---

35. Nonlinear Autoassociation Is Not Equivalent to PCA

Nathalie Japkowicz, Stephen José Hanson, Mark A. Gluck

https://doi.org/10.1162/089976600300015691 | pp. 531-545

A common misperception within the neural network community is that even with nonlinearities in their hidden layer, autoassociators trained with backpropagation are equivalent to linear methods such as principal component analysis (PCA). Our purpose is to demonstrate that nonlinear autoassociators actually behave differently from linear methods and that they can outperform these methods when used for latent extraction, projection, and classification. While linear autoassociators emulate PCA, and thus exhibit a flat or unimodal reconstruction error surface, autoassociators with nonlinearities in their hidden layer learn domains by building error reconstruction surfaces that, depending on the task, contain multiple local valleys. This interpolation bias allows nonlinear autoassociators to represent appropriate classifications of nonlinear multimodal domains, in contrast to linear autoassociators, which are inappropriate for such tasks. In fact, autoassociators with hidden unit nonlinearities can be shown to perform nonlinear classification and nonlinear recognition.

---

36. No Free Lunch for Noise Prediction

Malik Magdon■Ismail
https://doi.org/10.1162/089976600300015709 | pp. 547-564

No-free-lunch theorems have shown that learning algorithms cannot be universally good. We show that no free funch exists for noise prediction as well. We show that when the noise is additive and the prior over target functions is uniform, a prior on the noise distribution cannot be updated, in the Bayesian sense, from any finite data set. We emphasize the importance of a prior over the target function in order to justify superior performance for learning systems.

---

37. Self-Organization of Symmetry Networks: Transformation Invariance from the Spontaneous Symmetry-Breaking Mechanism

C. J. S. Webber
https://doi.org/10.1162/089976600300015718 | pp. 565-596

Symmetry networks use permutation symmetries among synaptic weights to achieve transformation-invariant response. This article proposes a generic mechanism by which such symmetries can develop during unsupervised adaptation: it is shown analytically that spontaneous symmetry breaking can result in the discovery of unknown invariances of the data's probability distribution. It is proposed that a role of sparse coding is to facilitate the discovery of statistical invariances by this mechanism. It is demonstrated that the statistical dependences that exist between simple-cell-like threshold feature detectors, when exposed to temporally uncorrelated natural image data, can drive the development of complex-cell-like invariances, via single-cell Hebbian adaptation. A single learning rule can generate both simple-cell-like and complex-cell-like receptive fields.

---

38. Geometric Analysis of Population Rhythms in Synaptically Coupled Neuronal Networks

Jonathan E. Rubin, David Terman
https://doi.org/10.1162/089976600300015727 | pp. 597-645

We develop geometric dynamical systems methods to determine how various components contribute to a neuronal network's emergent population behaviors. The results clarify the multiple roles inhibition can play in producing different rhythms. Which rhythms arise depends on how inhibition interacts with intrinsic properties of the neurons; the nature of these interactions depends on the underlying architecture of the network. Our analysis demonstrates that fast inhibitory coupling may lead to synchronized rhythms if either the cells within the network or the architecture of the network is sufficiently complicated. This cannot occur in mutually coupled networks with basic cells; the geometric approach helps explain how additional network complexity allows for synchronized rhythms in the presence of fast inhibitory coupling. The networks and issues considered are motivated by recent models for thalamic oscillations. The analysis helps clarify the roles of various biophysical features, such as fast and slow inhibition, cortical inputs, and ionic conductances, in producing network behavior associated with the spindle sleep rhythm and with paroxysmal discharge rhythms. Transitions between these rhythms are also discussed.

---

39. An Accurate Measure of the Instantaneous Discharge Probability, with Application to Unitary Joint-Event Analysis

Quentin Pauluis, Stuart N. Baker

https://doi.org/10.1162/089976600300015736 | pp. 647-669

We present an estimate for the instantaneous discharge probability of a neurone, based on single-trial spike-train analysis. By detecting points where the neurone abruptly changes its firing rate and treating them specially, the method is able to achieve smooth estimates yet avoid the blurring of significant changes. This estimate of instantaneous discharge probability is then applied to the method of unitary event analysis. We show that the unitary event analysis as originally conceived is highly sensitive to firing-rate nonstationarities and covariations, but that it can be considerably improved if calculations of statistical significance use an instantaneous discharge probability instead of a firing-rate estimate based on averaging across multiple trials.

---

40. Impact of Correlated Inputs on the Output of the Integrate-and-Fire Model

Jianfeng Feng, David Brown

https://doi.org/10.1162/089976600300015745 | pp. 671-692

For the integrate-and-fire model with or without reversal potentials, we consider how correlated inputs affect the variability of cellular output. For both models, the variability of efferent spike trains measured by coefficient of variation (CV) of the interspike interval is a nondecreasing function of input correlation. When the correlation coefficient is greater than 0.09, the CV of the integrate-and-fire model without reversal potentials is always above 0.5, no matter how strong the inhibitory inputs. When the correlation coefficient is greater than 0.05, CV for the integrate-and-fire model with reversal potentials is always above 0. 5, independent of the strength of the inhibitory inputs. Under a given condition on correlation coefficients, we find that correlated Poisson processes can be decomposed into independent Poisson processes. We also develop a novel method to estimate the distribution density of the first passage time of the integrate-and-fire model.

---

41. Weak, Stochastic Temporal Correlation of Large-Scale Synaptic Input Is a Major Determinant of Neuronal Bandwidth

David M. Halliday

https://doi.org/10.1162/089976600300015754 | pp. 693-707

We determine the bandwidth of a model neurone to large-scale synaptic input by assessing the frequency response between the outputs of a two-cell simulation that share a percentage of the total synaptic input. For temporally uncorrelated inputs, a large percentage of common inputs are required before the output discharges of the two cells exhibit significant correlation. In contrast, a small percentage (5%) of the total synaptic input that involves stochastic spike trains that are weakly correlated over a broad range of frequencies exert a clear influence on the output discharge of both cells over this range of frequencies. Inputs that are weakly correlated at a single frequency induce correlation between the output discharges only at the frequency of correlation. The strength of temporal correlation required is sufficiently weak that analysis of a sample pair of input spike trains could fail to reveal the presence of correlated input. Weak

temporal correlation between inputs is therefore a major determinant of the transmission to the output discharge of frequencies present in the spike discharges of presynaptic inputs, and therefore of neural bandwidth.

---

42. Second-Order Learning Algorithm with Squared Penalty Term

Kazumi Saito, Ryohei Nakano

https://doi.org/10.1162/089976600300015763 | pp. 709-729

This article compares three penalty terms with respect to the efficiency of supervised learning, by using first- and second-order off-line learning algorithms and a first-order on-line algorithm. Our experiments showed that for a reasonably adequate penalty factor, the combination of the squared penalty term and the second-order learning algorithm drastically improves the convergence performance in comparison to the other combinations, at the same time bringing about excellent generalization performance. Moreover, in order to understand how differently each penalty term works, a function surface evaluation is described. Finally, we show how cross validation can be applied to find an optimal penalty factor.

---

43. Minimizing Binding Errors Using Learned Conjunctive Features

Bartlett W. Mel, József Fiser

https://doi.org/10.1162/089976600300015574 | pp. 731-762

We have studied some of the design trade-offs governing visual representations based on spatially invariant conjunctive feature detectors, with an emphasis on the susceptibility of such systems to false-positive recognition errors-Malsburg's classical binding problem. We begin by deriving an analytical model that makes explicit how recognition performance is affected by the number of objects that must be distinguished, the number of features included in the representation, the complexity of individual objects, and the clutter load, that is, the amount of visual material in the field of view in which multiple objects must be simultaneously recognized, independent of pose, and without explicit segmentation. Using the domain of text to model object recognition in cluttered scenes, we show that with corrections for the nonuniform probability and nonindependence of text features, the analytical model achieves good fits to measured recognition rates in simulations involving a wide range of clutter loads, word size, and feature counts. We then introduce a greedy algorithm for feature learning, derived from the analytical model, which grows a representation by choosing those conjunctive features that are most likely to distinguish objects from the cluttered backgrounds in which they are embedded. We show that the representations produced by this algorithm are compact, decorrelated, and heavily weighted toward features of low conjunctive order. Our results provide a more quantitative basis for understanding when spatially invariant conjunctive features can support unambiguous perception in multiobject scenes, and lead to several insights regarding the properties of visual representations optimized for specific recognition tasks.

---

44. The Multifractal Structure of Contrast Changes in Natural Images: From Sharp Edges to Textures

Antonio Turiel, Néstor Parga

https://doi.org/10.1162/089976600300015583 | pp. 763-793

We present a formalism that leads naturally to a hierarchical description of the different contrast structures in images, providing precise definitions of sharp edges and other texture components. Within this formalism, we achieve a decomposition of pixels of the image in sets, the fractal components of the image, such that each set contains only points characterized by a fixed strength of the singularity of the contrast gradient in its neighborhood. A crucial role in this description of images is played by the behavior of contrast differences under changes in scale. Contrary to naive scaling ideas where the image is thought to have uniform transformation properties (Field, 1987), each of these fractal components has its own transformation law and scaling exponents. A conjecture on their biological relevance is also given.

---

45. Exponential or Polynomial Learning Curves? Case-Based Studies

Hanzhong Gu, Haruhisa Takahashi

https://doi.org/10.1162/089976600300015592 | pp. 795-809

Learning curves exhibit a diversity of behaviors such as phase transition. However, the understanding of learning curves is still extremely limited, and existing theories can give the impression that without empirical studies (e.g., cross validation), one can probably do nothing more than qualitative interpretations. In this note, we propose a theory of learning curves based on the idea of reducing learning problems to hypothesis-testing ones. This theory provides a simple approach that is potentially useful for predicting and interpreting (a diversity of) learning curve behaviors qualitatively and quantitatively, and it applies to finite training sample size and finite learning machine and for learning situations not necessarily within the Bayesian framework. We illustrate the results by examining some exponential learning curve behaviors observed in Cohn and Tesauro (1992)'s experiment.

---

46. Training Feedforward Neural Networks with Gain Constraints

Eric Hartman

https://doi.org/10.1162/089976600300015600 | pp. 811-829

Inaccurate input-output gains (partial derivatives of outputs with respect to inputs) are common in neural network models when input variables are correlated or when data are incomplete or inaccurate. Accurate gains are essential for optimization, control, and other purposes. We develop and explore a method for training feedforward neural networks subject to inequality or equality-bound constraints on the gains of the learned mapping. Gain constraints are implemented as penalty terms added to the objective function, and training is done using gradient descent. Adaptive and robust procedures are devised for balancing the relative strengths of the various terms in the objective function, which is essential when the constraints are inconsistent with the data. The approach has the virtue that the model domain of validity can be extended via extrapolation training, which can dramatically improve generalization. The algorithm is demonstrated here on artificial and real-world problems with very good results and has been advantageously applied to dozens of models currently in commercial use.

---

47. Variational Learning for Switching State-Space Models

Zoubin Ghahramani, Geoffrey E. Hinton
https://doi.org/10.1162/089976600300015619 | pp. 831-864

We introduce a new statistical model for time series that iteratively segments data into regimes with approximately linear dynamics and learnsthe parameters of each of these linear regimes. This model combines and generalizes two of the most widely used stochastic time-series models -- hidden Markov models and linear dynamical systems -- and is closely related to models that are widely used in the control and econometrics literatures. It can also be derived by extending the mixture of experts neural network (Jacobs, Jordan, Nowlan, & Hinton, 1991) to its fully dynamical version, in which both expert and gating networks are recurrent. Inferring the posterior probabilities of the hidden states of this model is computationally intractable, and therefore the exact expectation maximization (EM) algorithm cannot be applied. However, we present a variational approximation that maximizes a lower bound on the log-likelihood and makes use of both the forward and backward recursions for hidden Markov models and the Kalman filter recursions for linear dynamical systems. We tested the algorithm on artificial data sets and a natural data set of respiration force from a patient with sleep apnea. The results suggest that variational approximations are a viable method for inference and learning in switching state-space models.

---

48. Retrieval Properties of a Hopfield Model with Random Asymmetric Interactions

Zhang Chengxiang, Chandan Dasgupta, Manoranjan P. Singh
https://doi.org/10.1162/089976600300015628 | pp. 865-880

The process of pattern retrieval in a Hopfield model in which a random antisymmetric component is added to the otherwise symmetric synaptic matrix is studied by computer simulations. The introduction of the anti-symmetric component is found to increase the fraction of random inputs that converge to the memory states. However, the size of the basin of attraction of a memory state does not show any significant change when asymmetry is introduced in the synaptic matrix. We show that this is due to the fact that the spurious fixed points, which are destabilized by the introduction of asymmetry, have very small basins of attraction. The convergence time to spurious fixed-point attractors increases faster than that for the memory states as the asymmetry parameter is increased. The possibility of convergence to spurious fixed points is greatly reduced if a suitable upper limit is set for the convergence time. This prescription works better if the synaptic matrix has an antisymmetric component.

---

49. On “Natural” Learning and Pruning in Multilayered Perceptrons

Tom Heskes
https://doi.org/10.1162/089976600300015637 | pp. 881-901

Several studies have shown that natural gradient descent for on-line learning is much more efficient than standard gradient descent. In this article, we derive natural gradients in a slightly different manner and discuss implications for batch-mode learning and pruning, linking them to existing algorithms such as Levenberg-Marquardt optimization and optimal brain surgeon. The Fisher matrix plays an important role in all these algorithms. The second half of the article discusses a layered approximation of the Fisher matrix specific to multilayered perceptrons. Using this approximation rather than the exact Fisher matrix, we arrive

at much faster "natural" learning algorithms and more robust pruning procedures.

---

50. Synthesis of Generalized Algorithms for the Fast Computation of Synaptic Conductances with Markov Kinetic Models in Large Network Simulations

Michèle Giugliano

https://doi.org/10.1162/089976600300015646 | pp. 903-931

Markovkinetic models constitute a powerful framework to analyze patch-clamp data from single-channel recordings and model the dynamics of ion conductances and synaptic transmission between neurons. In particular, the accurate simulation of a large number of synaptic inputs in wide-scale network models may result in a computationally highly demanding process. We present a generalized consolidating algorithm to simulate efficiently a large number of synaptic inputs of the same kind (excitatory or inhibitory), converging on an isopotential compartment, independently modeling each synaptic current by a generic n-state Markov model characterized by piece-wise constant transition probabilities. We extend our findings to a class of simplified phenomenological descriptions of synaptic transmission that incorporate higher-order dynamics, such as short-term facilitation, depression, and synaptic plasticity.

---

51. Hierarchical Bayesian Models for Regularization in Sequential Learning

Jordan Freitas, Mahesan Niranjan, Andrew H. Gee

https://doi.org/10.1162/089976600300015655 | pp. 933-953

We show that a hierarchical Bayesian modeling approach allows us to perform regularization in sequential learning. We identify three inference levels within this hierarchy: model selection, parameter estimation, and noise estimation. In environments where data arrive sequentially, techniques such as cross validation to achieve regularization or model selection are not possible. The Bayesian approach, with extended Kalman filtering at the parameter estimation level, allows for regularization within a minimum variance framework. A multilayer perceptron is used to generate the extended Kalman filter nonlinear measurements mapping. We describe several algorithms at the noise estimation level that allow us to implement on-line regularization. We also show the theoretical links between adaptive noise estimation in extended Kalman filtering, multiple adaptive learning rates, and multiple smoothing regularization coefficients.

---

52. Sequential Monte Carlo Methods to Train Neural Network Models

Jordan Freitas, Mahesan Niranjan, Andrew H. Gee, Arnaud Doucet

https://doi.org/10.1162/089976600300015664 | pp. 955-993

We discuss a novel strategy for training neural networks using sequential Monte Carlo algorithms and propose a new hybrid gradient descent sampling importance resampling algorithm (HySIR). In terms of computational time and accuracy, the hybrid SIR is a clear improvement over conventional sequential Monte Carlo techniques. The new algorithm may be viewed as a global optimization strategy that allows us to learn the probability distributions of the network weights and outputs in a sequential framework. It is well suited to applications involving on-line, nonlinear, and nongaussian signal processing. We show how the new

algorithm outperforms extended Kalman filter training on several problems. In particular, we address the problem of pricing option contracts, traded in financial markets. In this context, we are able to estimate the one-step-ahead probability density functions of the options prices.

---

53. Expanding NEURON's Repertoire of Mechanisms with NMODL

Michael L. Hines, Nicholas T. Carnevale

https://doi.org/10.1162/089976600300015475 | pp. 995-1007

Neuronal function involves the interaction of electrical and chemical signals that are distributed in time and space. The mechanisms that generate these signals and regulate their interactions are marked by a rich diversity of properties that precludes a "one size fits all" approach to modeling. This article presents a summary of how the model description language NMODL enables the neuronal simulation environment NEURON to accommodate these differences.

---

54. Latent Attractors: A Model for Context-Dependent Place Representations in the Hippocampus

Simona Doboli, Ali A. Minai, Phillip J. Best

https://doi.org/10.1162/089976600300015484 | pp. 1009-1043

Cells throughout the rodent hippocampal system show place-specific patterns of firing called place fields, creating a coarse-coded representation of location. The dependencies of this place code--or cognitive map--on sensory cues have been investigated extensively, and several computational models have been developed to explain them. However, place representations also exhibit strong dependence on spatial and behavioral context, and identical sensory environments can produce very different place codes in different situations. Several recent studies have proposed models for the computational basis of this phenomenon, but it is still not completely understood. In this article, we present a very simple connectionist model for producing context-dependent place representations in the hippocampus. We propose that context dependence arises in the dentate gyrus-hilus (DGH) system, which functions as a dynamic selector, disposing a small group of granule and pyramidal cells to fire in response to afferent stimulus while depressing the rest. It is hypothesized that the DGH system dynamics has "latent attractors," which are unmasked by the afferent input and channel system activity into subpopulations of cells in the DG, CA3, and other hippocampal regions as observed experimentally. The proposed model shows that a minimally structured hippocampus-like system can robustly produce context-dependent place codes with realistic attributes.

---

55. The Approach of a Neuron Population Firing Rate to a New Equilibrium: An Exact Theoretical Result

B. W. Knight, Ahmet Omurtag, L. Sirovich

https://doi.org/10.1162/089976600300015493 | pp. 1045-1055

The response of a noninteracting population of identical neurons to a step change in steady synaptic input can be analytically calculated exactly from the dynamical equation that describes the population's evolution

in time. Here, for model integrate-and-fire neurons that undergo a fixed finite upward shift in voltage in response to each synaptic event, we compare the theoretical prediction with the result of a direct simulation of 90,000 model neurons. The degree of agreement supports the applicability of the population dynamics equation. The theoretical prediction is in the form of a series. Convergence is rapid, so that the full result is well approximated by a few terms.

---

56. Formation of Direction Selectivity in Natural Scene Environments

Brian S. Blais, Leon N. Cooper, Harel Z. Shouval

https://doi.org/10.1162/089976600300015501 | pp. 1057-1066

Most simple and complex cells in the cat striate cortex are both orientation and direction selective. In this article we use single-cell learning rules to develop both orientation and direction selectivity in a natural scene environment. We show that a simple principal component analysis rule is inadequate for developing direction selectivity, but that the BCM rule as well as similar higher-order rules can. We also demonstrate that the convergence of lagged and nonlagged cells depends on the velocity of motion in the environment, and that strobe rearing disrupts this convergence, resulting in a loss of direction selectivity.

---

57. A Phase Model of Temperature-Dependent Mammalian Cold Receptors

Peter Roper, Paul C. Bressloff, André Longtin

https://doi.org/10.1162/089976600300015510 | pp. 1067-1093

We present a tractable stochastic phase model of the temperature sensitivity of a mammalian cold receptor. Using simple linear dependencies of the amplitude, frequency, and bias on temperature, the model reproduces the experimentally observed transitions between bursting, beating, and stochastically phase-locked firing patterns. We analyze the model in the deterministic limit and predict, using a Strutt map, the number of spikes per burst for a given temperature. The inclusion of noise produces a variable number of spikes per burst and also extends the dynamic range of the neuron, both of which are analyzed in terms of the Strutt map. Our analysis can be readily applied to other receptors that display various bursting patterns following temperature changes.

---

58. The Number of Synaptic Inputs and the Synchrony of Large, Sparse Neuronal Networks

David Golomb, D. Hansel

https://doi.org/10.1162/089976600300015529 | pp. 1095-1139

The prevalence of coherent oscillations in various frequency ranges in the central nervous system raises the question of the mechanisms that synchronize large populations of neurons. We study synchronization in models of large networks of spiking neurons with random sparse connectivity. Synchrony occurs only when the average number of synapses, M, that a cell receives is larger than a critical value, Mc. Below Mc, the system is in an asynchronous state. In the limit of weak coupling, assuming identical neurons, we reduce the model to a system of phase oscillators that are coupled via an effective interaction, gamma. In this framework, we develop an approximate theory for sparse networks of identical neurons to estimate Mc

analytically from the Fourier coefficients of gamma. Our approach relies on the assumption that the dynamics of a neuron depend mainly on the number of cells that are presynaptic to it. We apply this theory to compute Mc for a model of inhibitory networks of integrate-and-fire (I&F) neurons as a function of the intrinsic neuronal properties (e.g., the refractory period Tr), the synaptic time constants, and the strength of the external stimulus, Iext. The number Mc is found to be nonmonotonous with the strength of Iext. For Tr = 0, we estimate the minimum value of Mc over all the parameters of the model to be 363.8. Above Mc, the neurons tend to fire in smeared one-cluster states at high firing rates and smeared two-or-more-cluster states at low firing rates. Refractoriness decreases Mc at intermediate and high firing rates. These results are compared to numerical simulations. We show numerically that systems with different sizes, N, behave in the same way provided the connectivity, M, is such that 1/Meff = 1/M - 1/N remains constant when N varies. This allows extrapolating the large N behavior of a network from numerical simulations of networks of relatively small sizes (N = 800 in our case). We find that our theory predicts with remarkable accuracy the value of Mc and the patterns of synchrony above Mc, provided the synaptic coupling is not too large. We also study the strong coupling regime of inhibitory sparse networks. All of our simulations demonstrate that increasing the coupling strength reduces the level of synchrony of the neuronal activity. Above a critical coupling strength, the network activity is asynchronous. We point out a fundamental limitation for the mechanisms of synchrony relying on inhibition alone, if heterogeneities in the intrinsic properties of the neurons and spatial fluctuations in the external input are also taken into account.

---

59. A Neural Network Architecture for Visual Selection

Yali Amit

https://doi.org/10.1162/089976600300015538 | pp. 1141-1164

This article describes a parallel neural net architecture for efficient and robust visual selection in generic gray-level images. Objects are represented through flexible star-type planar arrangements of binary local features which are in turn star-type planar arrangements of oriented edges. Candidate locations are detected over a range of scales and other deformations, using a generalized Hough transform. The flexibility of the arrangements provides the required invariance. Training involves selecting a small number of stable local features from a predefined pool, which are well localized on registered examples of the object. Training therefore requires only small data sets. The parallel architecture is constructed so that the Hough transform associated with any object can be implemented without creating or modifying any connections. The different object representations are learned and stored in a central module. When one of these representations is evoked, it "primes" the appropriate layers in the network so that the corresponding Hough transform is computed. Analogies between the different layers in the network and those in the visual system are discussed. Furthermore, the model can be used to explain certain experiments on visual selection reported in the literature.

---

60. Using Bayes' Rule to Model Multisensory Enhancement in the Superior Colliculus

Thomas J. Anastasio, Paul Patton, Kamel Belkacem-Boussaid

https://doi.org/10.1162/089976600300015547 | pp. 1165-1187

The deep layers of the superior colliculus (SC) integrate multisensory inputs and initiate an orienting response toward the source of stimulation (target). Multisensory enhancement, which occurs in the deep SC, is the augmentation of a neural response to sensory input of one modality by input of another modality. Multisensory enhancement appears to underlie the behavioral observation that an animal is more likely to orient toward weak stimuli if a stimulus of one modality is paired with a stimulus of another modality. Yet not all deep SC neurons are multisensory. Those that are exhibit the property of inverse effectiveness: combinations of weaker unimodal responses produce larger amounts of enhancement. We show that these neurophysiological findings support the hypothesis that deep SC neurons use their sensory inputs to compute the probability that a target is present. We model multimodal sensory inputs to the deep SC as random variables and cast the computation function in terms of Bayes' rule. Our analysis suggests that multisensory deep SC neurons are those that combine unimodal inputs that would be more uncertain by themselves. It also suggests that inverse effectiveness results because the increase in target probability due to the integration of multisensory inputs is larger when the unimodal responses are weaker.

---

61. Choice and Value Flexibility Jointly Contribute to the Capacity of a Subsampled Quadratic Classifier

Panayiota Poirazi, Bartlett W. Mel

https://doi.org/10.1162/089976600300015556 | pp. 1189-1205

Biophysical modeling studies have suggested that neurons with active dendrites can be viewed as linear units augmented by product terms that arise from interactions between synaptic inputs within the same dendritic subregions. However, the degree to which local nonlinear synaptic interactions could augment the memory capacity of a neuron is not known in a quantitative sense. To approach this question, we have studied the family of subsampled quadratic classifiers: linear classifiers augmented by the best k terms from the set of K = (d2 + d)/2 second-order product terms available in d dimensions. We developed an expression for the total parameter entropy, whose form shows that the capacity of an SQ classifier does not reside solely in its conventional weight values-the explicit memory used to store constant, linear, and higher-order coefficients. Rather, we identify a second type of parameter flexibility that jointly contributes to an SQ classifier's capacity: the choice as to which product terms are included in the model and which are not. We validate the form of the entropy expression using empirical studies of relative capacity within families of geometrically isomorphic SQ classifiers. Our results have direct implications for neurobiological (and other hardware) learning systems, where in the limit of high-dimensional input spaces and low-resolution synaptic weight values, this relatively little explored form of choice flexibility could constitute a major source of trainable model capacity.

---

62. New Support Vector Algorithms

Bernhard Schölkopf, Alex Smola, Robert C. Williamson, Peter L. Bartlett

https://doi.org/10.1162/089976600300015565 | pp. 1207-1245

We propose a new class of support vector algorithms for regression and classification. In these algorithms, a parameter nu lets one effectively control the number of support vectors. While this can be useful in its own right, the parameterization has the additional benefit of enabling us to eliminate one of the other free parameters of the algorithm: the accuracy parameter epsilon in the regression case, and the regularization

constant C in the classification case. We describe the algorithms, give some theoretical results concerning the meaning and the choice of nu, and report experimental results.

---

63. Separating Style and Content with Bilinear Models

Joshua B. Tenenbaum, William T. Freeman

https://doi.org/10.1162/089976600300015349 | pp. 1247-1283

Perceptual systems routinely separate "content" from "style," classifying familiar words spoken in an unfamiliar accent, identifying a font or handwriting style across letters, or recognizing a familiar face or object seen under unfamiliar viewing conditions. Yet a general and tractable computational model of this ability to untangle the underlying factors of perceptual observations remains elusive (Hofstadter, 1985). Existing factor models (Mardia, Kent, & Bibby, 1979; Hinton & Zemel, 1994; Ghahramani, 1995; Bell & Sejnowski, 1995; Hinton, Dayan, Frey, & Neal, 1995; Dayan, Hinton, Neal, & Zemel, 1995; Hinton & Ghahramani, 1997) are either insufficiently rich to capture the complex interactions of perceptually meaningful factors such as phoneme and speaker accent or letter and font, or do not allow efficient learning algorithms. We present a general framework for learning to solve two-factor tasks using bilinear models, which provide sufficiently expressive representations of factor interactions but can nonetheless be fit to data using efficient algorithms based on the singular value decomposition and expectation-maximization. We report promising results on three different tasks in three different perceptual domains: spoken vowel classification with a benchmark multi-speaker database, extrapolation of fonts to unseen letters, and translation of faces to novel illuminants.

---

64. Relationships Between the A Priori and A Posteriori Errors in Nonlinear Adaptive Neural Filters

Danilo P. Mandic, Jonathon A. Chambers

https://doi.org/10.1162/089976600300015358 | pp. 1285-1292

The lower bounds for the a posteriori prediction error of a nonlinear predictor realized as a neural network are provided. These are obtained for a priori adaptation and a posteriori error networks with sigmoid nonlinearities trained by gradient-descent learning algorithms. A contractivity condition is imposed on a nonlinear activation function of a neuron so that the a posteriori prediction error is smaller in magnitude than the corresponding a priori one. Furthermore, an upper bound is imposed on the learning rate eta so that the approach is feasible. The analysis is undertaken for both feedforward and recurrent nonlinear predictors realized as neural networks.

---

65. The VC Dimension for Mixtures of Binary Classifiers

Wenxin Jiang

https://doi.org/10.1162/089976600300015367 | pp. 1293-1301

The mixtures-of-experts (ME) methodology provides a tool of classification when experts of logistic regression models or Bernoulli models are mixed according to a set of local weights. We show that the Vapnik-Chervonenkis dimension of the ME architecture is bounded below by the number of experts m and above by O(m4s2), where s is the dimension of the input. For mixtures of Bernoulli experts with a scalar

input, we show that the lower bound m is attained, in which case we obtain the exact result that the VC dimension is equal to the number of experts.

---

66. The Early Restart Algorithm

Malik Magdon■Ismail, Amir F. Atiya

https://doi.org/10.1162/089976600300015376 | pp. 1303-1312

Consider an algorithm whose time to convergence is unknown (because of some random element in the algorithm, such as a random initial weight choice for neural network training). Consider the following strategy. Run the algorithm for a specific time T. If it has not converged by time T, cut the run short and rerun it from the start (repeat the same strategy for every run). This so-called restart mechanism has been proposed by Fahlman (1988) in the context of backpropagation training. It is advantageous in problems that are prone to local minima or when there is a large variability in convergence time from run to run, and may lead to a speed-up in such cases. In this article, we analyze theoretically the restart mechanism, and obtain conditions on the probability density of the convergence time for which restart will improve the expected convergence time. We also derive the optimal restart time. We apply the derived formulas to several cases, including steepest-descent algorithms.

---

67. Attractor Dynamics in Feedforward Neural Networks

Lawrence K. Saul, Michael I. Jordan

https://doi.org/10.1162/089976600300015385 | pp. 1313-1335

We study the probabilistic generative models parameterized by feedforward neural networks. An attractor dynamics for probabilistic inference in these models is derived from a mean field approximation for large, layered sigmoidal networks. Fixed points of the dynamics correspond to solutions of the mean field equations, which relate the statistics of each unit to those of its Markov blanket. We establish global convergence of the dynamics by providing a Lyapunov function and show that the dynamics generate the signals required for unsupervised learning. Our results for feedforward networks provide a counterpart to those of Cohen-Grossberg and Hopfield for symmetric networks.

---

68. Visualizing the Function Computed by a Feedforward Neural Network

Tony Plate, Joel L. Bert, John Grace, Pierre Band

https://doi.org/10.1162/089976600300015394 | pp. 1337-1353

A method for visualizing the function computed by a feedforward neural network is presented. It is most suitable for models with continuous inputs and a small number of outputs, where the output function is reasonably smooth, as in regression and probabilistic classification tasks. The visualization makes readily apparent the effects of each input and the way in which the functions deviate from a linear function. The visualization can also assist in identifying interactions in the fitted model. The method uses only the input-output relationship and thus can be applied to any predictive statistical model, including bagged and committee models, which are otherwise difficult to interpret. The visualization method is demonstrated on a neural network model of how the risk of lung cancer is affected by smoking and drinking.

69. Discriminant Pattern Recognition Using Transformation-Invariant Neurons

Diego Sona, Alessandro Sperduti, Antonina Starita

https://doi.org/10.1162/089976600300015402 | pp. 1355-1370

To overcome the problem of invariant pattern recognition, Simard, LeCun, and Denker (1993) proposed a successful nearest-neighbor approach based on tangent distance, attaining state-of-the-art accuracy. Since this approach needs great computational and memory effort, Hastie, Simard, and Säckinger (1995) proposed an algorithm (HSS) based on singular value decomposition (SVD), for the generation of nondiscriminant tangent models. In this article we propose a different approach, based on a gradient-descent constructive algorithm, called TD-Neuron, that develops discriminant models. We present as well comparative results of our constructive algorithm versus HSS and learning vector quantization (LVQ) algorithms. Specifically, we tested the HSS algorithm using both the original version based on the two-sided tangent distance and a new version based on the one-sided tangent distance. Empirical results over the NIST-3 database show that the TD-Neuron is superior to both SVD- and LVQ-based algorithms, since it reaches a better trade-off between error and rejection.

70. Observable Operator Models for Discrete Stochastic Time Series

Herbert Jaeger

https://doi.org/10.1162/089976600300015411 | pp. 1371-1398

A widely used class of models for stochastic systems is hidden Markov models. Systems that can be modeled by hidden Markov models are a proper subclass of linearly dependent processes, a class of stochastic systems known from mathematical investigations carried out over the past four decades. This article provides a novel, simple characterization of linearly dependent processes, called observable operator models. The mathematical properties of observable operator models lead to a constructive learning algorithm for the identification of linearly dependent processes. The core of the algorithm has a time complexity of O(N + nm3), where N is the size of training data, n is the number of distinguishable outcomes of observations, and m is model state-space dimension.

71. Adaptive Method of Realizing Natural Gradient Learning for Multilayer Perceptrons

■■■-■■■ ■■■■■*, Hyeyoung Park, Kenji Fukumizu*

https://doi.org/10.1162/089976600300015420 | pp. 1399-1409

The natural gradient learning method is known to have ideal performances for on-line training of multilayer perceptrons. It avoids plateaus, which give rise to slow convergence of the backpropagation method. It is Fisher efficient, whereas the conventional method is not. However, for implementing the method, it is necessary to calculate the Fisher information matrix and its inverse, which is practically very difficult. This article proposes an adaptive method of directly obtaining the inverse of the Fisher information matrix. It generalizes the adaptive Gauss-Newton algorithms and provides a solid theoretical justification of them. Simulations show that the proposed adaptive method works very well for realizing natural gradient learning.

72. Nonmonotonic Generalization Bias of Gaussian Mixture Models

Shotaro Akaho, Hilbert J. Kappen

https://doi.org/10.1162/089976600300015439 | pp. 1411-1427

Theories of learning and generalization hold that the generalization bias, defined as the difference between the training error and the generalization error, increases on average with the number of adaptive parameters. This article, however, shows that this general tendency is violated for a gaussian mixture model. For temperatures just below the first symmetry breaking point, the effective number of adaptive parameters increases and the generalization bias decreases. We compute the dependence of the neural information criterion on temperature around the symmetry breaking. Our results are confirmed by numerical cross-validation experiments.

73. Efficient Block Training of Multilayer Perceptrons

A. Navia-Vázquez, An■■bal R. Figueiras-Vidal

https://doi.org/10.1162/089976600300015448 | pp. 1429-1447

The attractive possibility of applying layerwise block training algorithms to multilayer perceptrons MLP, which offers initial advantages in computational effort, is refined in this article by means of introducing a sensitivity correction factor in the formulation. This results in a clear performance advantage, which we verify in several applications. The reasons for this advantage are discussed and related to implicit relations with second-order techniques, natural gradient formulations through Fisher's information matrix, and sample selection. Extensions to recurrent networks and other research lines are suggested at the close of the article.

74. An Optimization Approach to Design of Generalized BSB Neural Associative Memories

Jooyoung Park, Yonmook Park

https://doi.org/10.1162/089976600300015457 | pp. 1449-1462

This article is concerned with the synthesis of the optimally performing GBSB (generalized brain-state-in-a-box) neural associative memory given a set of desired binary patterns to be stored as asymptotically stable equilibrium points. Based on some known qualitative properties and newly observed fundamental properties of the GBSB model, the synthesis problem is formulated as a constrained optimization problem. Next, we convert this problem into a quasi-convex optimization problem called GEVP (generalized eigenvalue problem). This conversion is particularly useful in practice, because GEVPs can be efficiently solved by recently developed interior point methods. Design examples are given to illustrate the proposed approach and to compare with existing synthesis methods.

75. Nonholonomic Orthogonal Learning Algorithms for Blind Source Separation

■■■-■■■ ■■■■■, *Tianping Chen, Andrzej Cichocki*
https://doi.org/10.1162/089976600300015466 | pp. 1463-1484

Independent component analysis or blind source separation extracts independent signals from their linear mixtures without assuming prior knowledge of their mixing coefficients. It is known that the independent signals in the observed mixtures can be successfully extracted except for their order and scales. In order to resolve the indeterminacy of scales, most learning algorithms impose some constraints on the magnitudes of the recovered signals. However, when the source signals are nonstationary and their average magnitudes change rapidly, the constraints force a rapid change in the magnitude of the separating matrix. This is the case with most applications (e.g., speech sounds, electroencephalogram signals). It is known that this causes numerical instability in some cases. In order to resolve this difficulty, this article introduces new nonholonomic constraints in the learning algorithm. This is motivated by the geometrical consideration that the directions of change in the separating matrix should be orthogonal to the equivalence class of separating matrices due to the scaling indeterminacy. These constraints are proved to be nonholonomic, so that the proposed algorithm is able to adapt to rapid or intermittent changes in the magnitudes of the source signals. The proposed algorithm works well even when the number of the sources is overestimated, whereas the existent algorithms do not (assuming the sensor noise is negligibly small), because they amplify the null components not included in the sources. Computer simulations confirm this desirable property.

---

76. The Minimal Local-Asperity Hypothesis of Early Retinal Lateral Inhibition

Rosario M. Balboa, Norberto M. Grzywacz
https://doi.org/10.1162/089976600300015231 | pp. 1485-1517

Recently we found that the theories related to information theory existent in the literature cannot explain the behavior of the extent of the lateral inhibition mediated by retinal horizontal cells as a function of background light intensity. These theories can explain the fall of the extent from intermediate to high intensities, but not its rise from dim to intermediate intensities. We propose an alternate hypothesis that accounts for the extent's bell-shape behavior. This hypothesis proposes that the lateral-inhibition adaptation in the early retina is part of a system to extract several image attributes, such as occlusion borders and contrast. To do so, this system would use prior probabilistic knowledge about the biological processing and relevant statistics in natural images. A key novel statistic used here is the probability of the presence of an occlusion border as a function of local contrast. Using this probabilistic knowledge, the retina would optimize the spatial profile of lateral inhibition to minimize attribute-extraction error. The two significant errors that this minimization process must reduce are due to the quantal noise in photoreceptors and the straddling of occlusion borders by lateral inhibition.

---

77. Multidimensional Encoding Strategy of Spiking Neurons

Christian W. Eurich, Stefan Wilke
https://doi.org/10.1162/089976600300015240 | pp. 1519-1529

Neural responses in sensory systems are typically triggered by a multitude of stimulus features. Using information theory, we study the encoding accuracy of a population of stochastically spiking neurons characterized by different tuning widths for the different features. The optimal encoding strategy for representing one feature most accurately consists of narrow tuning in the dimension to be encoded, to

increase the single-neuron Fisher information, and broad tuning in all other dimensions, to increase the number of active neurons. Extremely narrow tuning without sufficient receptive field overlap will severely worsen the coding. This implies the existence of an optimal tuning width for the feature to be encoded. Empirically, only a subset of all stimulus features will normally be accessible. In this case, relative encoding errors can be calculated that yield a criterion for the function of a neural population based on the measured tuning curves.

---

78. Synergy in a Neural Code

Naama Brenner, Steven Strong, R. Köberle, William Bialek, Rob R. de Ruyter van Steveninck

https://doi.org/10.1162/089976600300015259 | pp. 1531-1552

We show that the information carried by compound events in neural spike trains-patterns of spikes across time or across a population of cells-can be measured, independent of assumptions about what these patterns might represent. By comparing the information carried by a compound pattern with the information carried independently by its parts, we directly measure the synergy among these parts. We illustrate the use of these methods by applying them to experiments on the motion-sensitive neuron H1 of the fly's visual system, where we confirm that two spikes close together in time carry far more than twice the information carried by a single spike. We analyze the sources of this synergy and provide evidence that pairs of spikes close together in time may be especially important patterns in the code of H1.

---

79. Increased Synchrony with Increase of a Low-Threshold Calcium Conductance in a Model Thalamic Network: A Phase-Shift Mechanism

Elizabeth Thomas, Thierry Grisar

https://doi.org/10.1162/089976600300015268 | pp. 1553-1571

A computer model of a thalamic network was used in order to examine the effects of an isolated augmentation in a low-threshold calcium current. Such an isolated augmentation has been observed in the reticular thalamic (RE) nucleus of the genetic absence epilepsy rat from the Strasbourg (GAERS) model of absence epilepsy. An augmentation of the low-threshold calcium conductance in the RE neurons (gTs) of the model thalamic network was found to lead to an increase in the synchronized firing of the network. This supports the hypothesis that the isolated increase in gTs may be responsible for epileptic activity in the GAERS rat. The increase of gTs in the RE neurons led to a slight increase in the period of the isolated RE neuron firing. In contrast, the low-threshold spike of the RE neuron remained relatively unchanged by the increase of gTs. This suggests that the enhanced synchrony in the network was primarily due to a phase shift in the firing of the RE neurons with respect to the thalamocortical neurons. The ability of this phase-shift mechanism to lead to changes in synchrony was further examined using the model thalamic network. A similar increase in the period of RE neuron oscillations was obtained through an increase in the conductance of the calcium-mediated potassium channel. This change was once again found to increase synchronous firing in the network.

---

80. Multispikes and Synchronization in a Large Neural Network with Temporal Delays

Jan Karbowski, Nancy Kopell

https://doi.org/10.1162/089976600300015277 | pp. 1573-1606

Coherent rhythms in the gamma frequency range are ubiquitous in the nervous system and thought to be important in a variety of cognitive activities. Such rhythms are known to be able to synchronize with millisecond precision across distances with significant conduction delay; it is mysterious how this can operate in a setting in which cells receive many inputs over a range of time. Here we analyze a version of mechanism, previously proposed, that the synchronization in the CA1 region of the hippocampus depends on the firing of "doublets" by the interneurons. Using a network of local circuits that are arranged in a possibly disordered lattice, we determine the conditions on parameters for existence and stability of synchronous solutions in which the inhibitory interneurons fire single spikes, doublets, or triplets per cycle. We show that the synchronous solution is only marginally stable if the interneurons fire singlets. If they fire doublets, the synchronous state is asymptotically stable in a larger subset of parameter space than if they fire triplets. An unexpected finding is that a small amount of disorder in the lattice structure enlarges the parameter regime in which the doublet solution is stable. Synaptic noise reduces the regime in which the doublet configuration is stable, but only weakly.

---

81. Synchrony in Heterogeneous Networks of Spiking Neurons

L. Neltner, David Hansel, Germán Mato, Claude Meunier

https://doi.org/10.1162/089976600300015286 | pp. 1607-1641

The emergence of synchrony in the activity of large, heterogeneous networks of spiking neurons is investigated. We define the robustness of synchrony by the critical disorder at which the asynchronous state becomes linearly unstable. We show that at low firing rates, synchrony is more robust in excitatory networks than in inhibitory networks, but excitatory networks cannot display any synchrony when the average firing rate becomes too high. We introduce a new regime where all inputs, external and internal, are strong and have opposite effects that cancel each other when averaged. In this regime, the robustness of synchrony is strongly enhanced, and robust synchrony can be achieved at a high firing rate in inhibitory networks. On the other hand, in excitatory networks, synchrony remains limited in frequency due to the intrinsic instability of strong recurrent excitation.

---

82. Dynamics of Spiking Neurons with Electrical Coupling

Carson C. Chow, Nancy Kopell

https://doi.org/10.1162/089976600300015295 | pp. 1643-1678

We analyze the existence and stability of phase-locked states of neurons coupled electrically with gap junctions. We show that spike shape and size, along with driving current (which affects network frequency), play a large role in which phase-locked modes exist and are stable. Our theory makes predictions about biophysical models using spikes of different shapes, and we present simulations to confirm the predictions. We also analyze a large system of all-to-all coupled neurons and show that the splay-phase state can exist only for a certain range of frequencies.

83. A Model for Fast Analog Computation Based on Unreliable Synapses

Wolfgang Maass, Thomas Natschläger

https://doi.org/10.1162/089976600300015303 | pp. 1679-1704

We investigate through theoretical analysis and computer simulations the consequences of unreliable synapses for fast analog computations in networks of spiking neurons, with analog variables encoded by the current firing activities of pools of spiking neurons. Our results suggest a possible functional role for the well-established unreliability of synaptic transmission on the network level. We also investigate computations on time series and Hebbian learning in this context of space-rate coding in networks of spiking neurons with unreliable synapses.

84. Emergence of Phase- and Shift-Invariant Features by Decomposition of Natural Images into Independent Feature Subspaces

Aapo Hyvärinen, Patrik O. Hoyer

https://doi.org/10.1162/089976600300015312 | pp. 1705-1720

Olshausen and Field (1996) applied the principle of independence maximization by sparse coding to extract features from natural images. This leads to the emergence of oriented linear filters that have simultaneous localization in space and in frequency, thus resembling Gabor functions and simple cell receptive fields. In this article, we show that the same principle of independence maximization can explain the emergence of phase- and shift-invariant features, similar to those found in complex cells. This new kind of emergence is obtained by maximizing the independence between norms of projections on linear subspaces (instead of the independence of simple linear filter outputs). The norms of the projections on such "independent feature subspaces" then indicate the values of invariant features.

85. Tilt Aftereffects in a Self-Organizing Model of the Primary Visual Cortex

James A. Bednar, Risto Miikkulainen

https://doi.org/10.1162/089976600300015321 | pp. 1721-1740

RF-LISSOM, a self-organizing model of laterally connected orientation maps in the primary visual cortex, was used to study the psychological phenomenon known as the tilt aftereffect. The same self-organizing processes that are responsible for the long-term development of the map are shown to result in tilt aftereffects over short timescales in the adult. The model permits simultaneous observation of large numbers of neurons and connections, making it possible to relate high-level phenomena to low-level events, which is difficult to do experimentally. The results give detailed computational support for the long-standing conjecture that the direct tilt aftereffect arises from adaptive lateral interactions between feature detectors. They also make a new prediction that the indirect effect results from the normalization of synaptic efficacies during this process. The model thus provides a unified computational explanation of self-organization and both the direct and indirect tilt aftereffect in the primary visual cortex.

86. Errata

(authors unknown)

https://doi.org/10.1162/089976600300015330 | pp. 1741-1741

No abstract available.

---

87. Neural Systems as Nonlinear Filters

Wolfgang Maass, Eduardo D. Sontag

https://doi.org/10.1162/089976600300015123 | pp. 1743-1772

Experimental data show that biological synapses behave quite differently from the symbolic synapses in all common artificial neural network models. Biological synapses are dynamic; their "weight" changes on a short timescale by several hundred percent in dependence of the past input to the synapse. In this article we address the question how this inherent synaptic dynamics (which should not be confused with long term learning) affects the computational power of a neural network. In particular, we analyze computations on temporal and spatiotemporal patterns, and we give a complete mathematical characterization of all filters that can be approximated by feedforward neural networks with dynamic synapses. It turns out that even with just a single hidden layer, such networks can approximate a very rich class of nonlinear filters: all filters that can be characterized by Volterra series. This result is robust with regard to various changes in the model for synaptic dynamics. Our characterization result provides for all nonlinear filters that are approximable by Volterra series a new complexity hierarchy related to the cost of implementing such filters in neural systems.

---

88. Analytical Model for the Effects of Learning on Spike Count Distributions

Giovanni Settanni, Alessandro Treves

https://doi.org/10.1162/089976600300015132 | pp. 1773-1787

The spike count distribution observed when recording from a variety of neurons in many different conditions has a fairly stereotypical shape, with a single mode at zero or close to a low average count, and a long, quasi-exponential tail to high counts. Such a distribution has been suggested to be the direct result of three simple facts: the firing frequency of a typical cortical neuron is close to linear in the summed input current entering the soma, above a threshold; the input current varies on several timescales, both faster and slower than the window used to count spikes; and the input distribution at any timescale can be taken to be approximately normal. The third assumption is violated by associative learning, which generates correlations between the synaptic weight vector on the dendritic tree of a neuron, and the input activity vectors it is repeatedly subject to. We show analytically that for a simple feed-forward model, the normal distribution of the slow components of the input current becomes the sum of two quasi-normal terms. The term important below threshold shifts with learning, while the term important above threshold does not shift but grows in width. These deviations from the standard distribution may be observable in appropriate recording experiments.

---

89. Calculation of Interspike Intervals for Integrate-and-Fire Neurons with Poisson Distribution of Synaptic Inputs

Anthony N. Burkitt, Graeme M. Clark
https://doi.org/10.1162/089976600300015141 | pp. 1789-1820

We present a new technique for calculating the interspike intervals of integrate-and-fire neurons. There are two new components to this technique. First, the probability density of the summed potential is calculated by integrating over the distribution of arrival times of the afferent post-synaptic potentials (PSPs), rather than using conventional stochastic differential equation techniques. A general formulation of this technique is given in terms of the probability distribution of the inputs and the time course of the postsynaptic response. The expressions are evaluated in the gaussian approximation, which gives results that become more accurate for large numbers of small-amplitude PSPs. Second, the probability density of output spikes, which are generated when the potential reaches threshold, is given in terms of an integral involving a conditional probability density. This expression is a generalization of the renewal equation, but it holds for both leaky neurons and situations in which there is no time-translational invariance. The conditional probability density of the potential is calculated using the same technique of integrating over the distribution of arrival times of the afferent PSPs. For inputs with a Poisson distribution, the known analytic solutions for both the perfect integrator model and the Stein model (which incorporates membrane potential leakage) in the diffusion limit are obtained. The interspike interval distribution may also be calculated numerically for models that incorporate both membrane potential leakage and a finite rise time of the postsynaptic response. Plots of the relationship between input and output firing rates, as well as the coefficient of variation, are given, and inputs with varying rates and amplitudes, including inhibitory inputs, are analyzed. The results indicate that neurons functioning near their critical threshold, where the inputs are just sufficient to cause firing, display a large variability in their spike timings.

---

90. Statistical Procedures for Spatiotemporal Neuronal Data with Applications to Optical Recording of the Auditory Cortex

Olivier François, L. M. Ould Mohamed Abdallahi, Junsei Horikawa, Ikuo Taniguchi, T. Hervé
https://doi.org/10.1162/089976600300015150 | pp. 1821-1838

This article presents new procedures for multisite spatiotemporal neuronal data analysis. A new statistical model - the diffusion model - is considered, whose parameters can be estimated from experimental data thanks to mean-field approximations. This work has been applied to optical recording of the guinea pig's auditory cortex (layers II-III). The rates of innovation and internal diffusion inside the stimulated area have been estimated. The results suggest that the activity of the layer balances between the alternate predominance of its innovation process and its internal process.

---

91. Probabilistic Motion Estimation Based on Temporal Coherence

Pierre■Yves Burgi, Alan Yuille, Norberto M. Grzywacz
https://doi.org/10.1162/089976600300015169 | pp. 1839-1867

We develop a theory for the temporal integration of visual motion motivated by psychophysical experiments. The theory proposes that input data are temporally grouped and used to predict and estimate the motion flows in the image sequence. This temporal grouping can be considered a generalization of the data association techniques that engineers use to study motion sequences. Our temporal grouping theory is

expressed in terms of the Bayesian generalization of standard Kalman filtering. To implement the theory, we derive a parallel network that shares some properties of cortical networks. Computer simulations of this network demonstrate that our theory qualitatively accounts for psychophysical experiments on motion occlusion and motion outliers. In deriving our theory, we assumed spatial factorizability of the probability distributions and made the approximation of updating the marginal distributions of velocity at each point. This allowed us to perform local computations and simplified our implementation. We argue that these approximations are suitable for the stimuli we are considering (for which spatial coherence effects are negligible).

---

92. Boosting Neural Networks

Holger Schwenk, Yoshua Bengio

https://doi.org/10.1162/089976600300015178 | pp. 1869-1887

Boosting is a general method for improving the performance of learning algorithms. A recently proposed boosting algorithm, AdaBoost, has been applied with great success to several benchmark machine learning problems using mainly decision trees as base classifiers. In this article we investigate whether AdaBoost also works as well with neural networks, and we discuss the advantages and drawbacks of different versions of the AdaBoost algorithm. In particular, we compare training methods based on sampling the training set and weighting the cost function. The results suggest that random resampling of the training data is not the main explanation of the success of the improvements brought by AdaBoost. This is in contrast to bagging, which directly aims at reducing variance and for which random resampling is essential to obtain the reduction in generalization error. Our system achieves about 1.4% error on a data set of on-line handwritten digits from more than 200 writers. A boosted multilayer network achieved 1.5% error on the UCI letters and 8.1% error on the UCI satellite data set, which is significantly better than boosted decision trees.

---

93. Gradient-Based Optimization of Hyperparameters

Yoshua Bengio

https://doi.org/10.1162/089976600300015187 | pp. 1889-1900

Many machine learning algorithms can be formulated as the minimization of a training criterion that involves a hyperparameter. This hyperparameter is usually chosen by trial and error with a model selection criterion. In this article we present a methodology to optimize several hyperparameters, based on the computation of the gradient of a model selection criterion with respect to the hyperparameters. In the case of a quadratic training criterion, the gradient of the selection criterion with respect to the hyperparameters is efficiently computed by backpropagating through a Cholesky decomposition. In the more general case, we show that the implicit function theorem can be used to derive a formula for the hyperparameter gradient involving second derivatives of the training criterion.

---

94. A Signal-Flow-Graph Approach to On-line Gradient Calculation

P. Campolucci, Aurelio Uncini, Francesco Piazza

https://doi.org/10.1162/089976600300015196 | pp. 1901-1927

A large class of nonlinear dynamic adaptive systems such as dynamic recurrent neural networks can be effectively represented by signal flow graphs (SFGs). By this method, complex systems are described as a general connection of many simple components, each of them implementing a simple one-input, one-output transformation, as in an electrical circuit. Even if graph representations are popular in the neural network community, they are often used for qualitative description rather than for rigorous representation and computational purposes. In this article, a method for both on-line and batch-backward gradient computation of a system output or cost function with respect to system parameters is derived by the SFG representation theory and its known properties. The system can be any causal, in general nonlinear and time-variant, dynamic system represented by an SFG, in particular any feedforward, time-delay, or recurrent neural network. In this work, we use discrete-time notation, but the same theory holds for the continuous-time case. The gradient is obtained in a straightforward way by the analysis of two SFGs, the original one and its adjoint (obtained from the first by simple transformations), without the complex chain rule expansions of derivatives usually employed. This method can be used for sensitivity analysis and for learning both off-line and on-line. On-line learning is particularly important since it is required by many real applications, such as digital signal processing, system identification and control, channel equalization, and predistortion.

---

95. Bootstrapping Neural Networks

Jürgen Franke, Michael H. Neumann

https://doi.org/10.1162/089976600300015204 | pp. 1929-1949

Knowledge about the distribution of a statistical estimator is important for various purposes, such as the construction of confidence intervals for model parameters or the determination of critical values of tests. A widely used method to estimate this distribution is the so-called boot-strap, which is based on an imitation of the probabilistic structure of the data-generating process on the basis of the information provided by a given set of random observations. In this article we investigate this classical method in the context of artificial neural networks used for estimating a mapping from input to output space. We establish consistency results for bootstrap estimates of the distribution of parameter estimates.

---

96. Information Geometry of Mean-Field Approximation

Toshiyuki Tanaka

https://doi.org/10.1162/089976600300015213 | pp. 1951-1968

I present a general theory of mean-field approximation based on information geometry and applicable not only to Boltzmann machines but also to wider classes of statistical models. Using perturbation expansion of the Kullback divergence (or Plefka expansion in statistical physics), a formulation of mean-field approximation of general orders is derived. It includes in a natural way the "naive" mean-field approximation and is consistent with the Thouless-Anderson-Palmer (TAP) approach and the linear response theorem in statistical physics.

---

97. Measuring the VC-Dimension Using Optimized Experimental Design

Xuhui Shao, Vladimir Cherkassky, William Li

https://doi.org/10.1162/089976600300015222 | pp. 1969-1986

VC-dimension is the measure of model complexity (capacity) used in VC-theory. The knowledge of the VC-dimension of an estimator is necessary for rigorous complexity control using analytic VC generalization bounds. Unfortunately, it is not possible to obtain the analytic estimates of the VC-dimension in most cases. Hence, a recent proposal is to measure the VC-dimension of an estimator experimentally by fitting the theoretical formula to a set of experimental measurements of the frequency of errors on artificially generated data sets of varying sizes (Vapnik, Levin, & Le Cun, 1994). However, it may be difficult to obtain an accurate estimate of the VC-dimension due to the variability of random samples in the experimental procedure proposed by Vapnik et al. (1994). We address this problem by proposing an improved design procedure for specifying the measurement points (i.e., the sample size and the number of repeated experiments at a given sample size). Our approach leads to a nonuniform design structure as opposed to the uniform design structure used in the original article (Vapnik et al., 1994). Our simulation results show that the proposed optimized design structure leads to a more accurate estimation of the VC-dimension using the experimental procedure. The results also show that a more accurate estimation of VC-dimension leads to improved complexity control using analytic VC-generalization bounds and, hence, better prediction accuracy.

---

98. DEA: An Architecture for Goal Planning and Classification

François Fleuret, Éric Brunet-Gouet

https://doi.org/10.1162/089976600300015024 | pp. 1987-2008

We introduce the differential efficiency algorithm, which partitions a perceptive space during unsupervised learning into categories and uses them to solve goal-planning and classification problems. This algorithm is inspired by a biological model of the cortex proposing the cortical column as an elementary unit. We validate the generality of this approach by testing it on four problems with continuous time and no reinforcement signal until the goal is reached (constrained object moves, Hanoi tower problem, animat control, and simple character recognition).

---

99. On a Fast, Compact Approximation of the Exponential Function

Gavin C. Cawley

https://doi.org/10.1162/089976600300015033 | pp. 2009-2012

Recently Schraudolph (1999) described an ingenious, fast, and compact approximation of the exponential function through manipulation of the components of a standard (IEEE-754 (IEEE, 1985)) floating-point representation. This brief note communicates a recoding of this procedure that overcomes some of the limitations of the original macro at little or no additional computational expense.

---

100. Bounds on Error Expectation for Support Vector Machines

Vladimir Vapnik, Olivier Chapelle

https://doi.org/10.1162/089976600300015042 | pp. 2013-2036

We introduce the concept of span of support vectors (SV) and show that the generalization ability of support vector machines (SVM) depends on this new geometrical concept. We prove that the value of the span is

always smaller (and can be much smaller) than the diameter of the smallest sphere containing the support vectors, used in previous bounds (Vapnik, 1998). We also demonstrate experimentally that the prediction of the test error given by the span is very accurate and has direct application in model selection (choice of the optimal parameters of the SVM).

---

101. A Computational Model of Lateralization and Asymmetries in Cortical Maps

Svetlana Levitan, James A. Reggia

https://doi.org/10.1162/089976600300015051 | pp. 2037-2062

While recent experimental work has defined asymmetries and lateralization in left and right cortical maps, the mechanisms underlying these phenomena are currently not established. In order to explore some possible mechanisms in theory, we studied a neural model consisting of paired cerebral hemispheric regions interacting via a simulated corpus callosum. Starting with random synaptic strengths, unsupervised (Hebbian) synaptic modifications led to the emergence of a topographic map in one or both hemispheric regions. Because of uncertainties concerning the nature of hemispheric interactions, both excitatory and inhibitory callosal influences were examined independently. A sharp transition in model behavior was observed depending on callosal strength. For excitatory or weakly inhibitory callosal interactions, complete and symmetric mirror-image maps generally appeared in both hemispheric regions. In contrast, with stronger inhibitory callosal interactions, partial to complete map lateralization tended to occur, and the maps in each hemispheric region often became complementary. Lateralization occurred readily toward the side having a larger cortical region or higher excitability. Asymmetric synaptic plasticity, however, had only a transitory effect on lateralization. These results support the hypotheses that interhemispheric competition occurs, that multiple underlying asymmetries may lead to function lateralization, and that the effects of asymmetric synaptic plasticity may vary depending on whether supervised or unsupervised learning is involved. To our knowledge, this is the first computational model to demonstrate the emergence of topographic map lateralization and asymmetries.

---

102. Rate Limitations of Unitary Event Analysis

Arup Roy, Peter N. Steinmetz, Ernst Niebur

https://doi.org/10.1162/089976600300015060 | pp. 2063-2082

Unitary event analysis is a new method for detecting episodes of synchronized neural activity (Riehle, Grün, Diesmann, & Aertsen, 1997). It detects time intervals that contain coincident firing at higher rates than would be expected if the neurons fired as independent inhomogeneous Poisson processes; all coincidences in such intervals are called unitary events (UEs). Changes in the frequency of UEs that are correlated with behavioral states may indicate synchronization of neural firing that mediates or represents the behavioral state. We show that UE analysis is subject to severe limitations due to the underlying discrete statistics of the number of coincident events. These limitations are particularly stringent for low (0-10 spikes/s) firing rates. Under these conditions, the frequency of UEs is a random variable with a large variation relative to its mean. The relative variation decreases with increasing firing rate, and we compute the lowest firing rate, at which the 95% confidence interval around the mean frequency of UEs excludes zero. This random variation in UE frequency makes interpretation of changes in UEs problematic for neurons with low firing rates. As a

typical example, when analyzing 150 trials of an experiment using an averaging window 100 ms wide and a 5 ms coincidence window, firing rates should be greater than 7 spikes per second.

---

103. Estimating Functions of Independent Component Analysis for Temporally Correlated Signals

■■■-■■■ ■■■■■
https://doi.org/10.1162/089976600300015079 | pp. 2083-2107

This article studies a general theory of estimating functions of independent component analysis when the independent source signals are temporarily correlated. Estimating functions are used for deriving both batch and on-line learning algorithms, and they are applicable to blind cases where spatial and temporal probability structures of the sources are unknown. Most algorithms proposed so far can be analyzed in the framework of estimating functions. An admissible class of estimating functions is derived, and related efficient on-line learning algorithms are introduced. We analyze dynamical stability and statistical efficiency of these algorithms. Different from the independently and identically distributed case, the algorithms work even when only the second-order moments are used. The method of simultaneous diagonalization of cross-covariance matrices is also studied from the point of view of estimating functions.

---

104. SMEM Algorithm for Mixture Models

Naonori Ueda, Ryohei Nakano, Zoubin Ghahramani, Geoffrey E. Hinton
https://doi.org/10.1162/089976600300015088 | pp. 2109-2128

We present a split-and-merge expectation-maximization (SMEM) algorithm to overcome the local maxima problem in parameter estimation of finite mixture models. In the case of mixture models, local maxima often involve having too many components of a mixture model in one part of the space and too few in another, widely separated part of the space. To escape from such configurations, we repeatedly perform simultaneous split-and-merge operations using a new criterion for efficiently selecting the split-and-merge candidates. We apply the proposed algorithm to the training of gaussian mixtures and mixtures of factor analyzers using synthetic and real data and show the effectiveness of using the split-and-merge operations to improve the likelihood of both the training data and of held-out test data. We also show the practical usefulness of the proposed algorithm by applying it to image compression and pattern recognition problems.

---

105. Stable Encoding of Finite-State Machines in Discrete-Time Recurrent Neural Nets with Sigmoid Units

Rafael C. Carrasco, Mikel L. Forcada, M. Ángeles Valdés-Muñoz, Ramón P. Ñeco
https://doi.org/10.1162/089976600300015097 | pp. 2129-2174

There has been a lot of interest in the use of discrete-time recurrent neural nets (DTRNN) to learn finite-state tasks, with interesting results regarding the induction of simple finite-state machines from input-output strings. Parallel work has studied the computational power of DTRNN in connection with finite-state computation. This article describes a simple strategy to devise stable encodings of finite-state machines in computationally capable discrete-time recurrent neural architectures with sigmoid units and

gives a detailed presentation on how this strategy may be applied to encode a general class of finite-state machines in a variety of commonly used first- and second-order recurrent neural networks. Unlike previous work that either imposed some restrictions to state values or used a detailed analysis based on fixed-point attractors, our approach applies to any positive, bounded, strictly growing, continuous activation function and uses simple bounding criteria based on a study of the conditions under which a proposed encoding scheme guarantees that the DTRNN is actually behaving as a finite-state machine.

---

106. Approximate Maximum Entropy Joint Feature Inference Consistent with Arbitrary Lower-Order Probability Constraints: Application to Statistical Classification

David J. Miller, Lian Yan

https://doi.org/10.1162/089976600300015105 | pp. 2175-2207

We propose a new learning method for discrete space statistical classifiers. Similar to Chow and Liu (1968) and Cheeseman (1983), we cast classification/inference within the more general framework of estimating the joint probability mass function (p.m.f.) for the (feature vector, class label) pair. Cheeseman's proposal to build the maximum entropy (ME) joint p.m.f. consistent with general lower-order probability constraints is in principle powerful, allowing general dependencies between features. However, enormous learning complexity has severely limited the use of this approach. Alternative models such as Bayesian networks (BNs) require explicit determination of conditional independencies. These may be difficult to assess given limited data. Here we propose an approximate ME method, which, like previous methods, incorporates general constraints while retaining quite tractable learning. The new method restricts joint p.m.f. support during learning to a small subset of the full feature space. Classification gains are realized over dependence trees, tree-augmented naive Bayes networks, BNs trained by the Kutato algorithm, and multilayer perceptrons. Extensions to more general inference problems are indicated. We also propose a novel exact inference method when there are several missing features.

---

107. λ-Opt Neural Approaches to Quadratic Assignment Problems

Shin Ishii, Hirotaka Niitsuma

https://doi.org/10.1162/089976600300015114 | pp. 2209-2225

In this article, we propose new analog neural approaches to combinatorial optimization problems, in particular, quadratic assignment problems (QAPs). Our proposed methods are based on an analog version of the lambda-opt heuristics, which simultaneously changes assignments for lambda elements in a permutation. Since we can take a relatively large lambda value, our new methods can achieve a middle-range search over possible solutions, and this helps the system neglect shallow local minima and escape from local minima. In experiments, we have applied our methods to relatively large-scale (N = 80 - 150) QAPs. Results have shown that our new methods are comparable to the present champion algorithms; for two benchmark problems, they are obtain better solutions than the previous champion algorithms.

---

108. Spike-Driven Synaptic Plasticity: Theory, Simulation, VLSI Implementation

Stefano Fusi, Mario Annunziato, D. Badoni, Á. Salamon, Daniel J. Amit

https://doi.org/10.1162/089976600300014917 | pp. 2227-2258

We present a model for spike-driven dynamics of a plastic synapse, suited for aVLSI implementation. The synaptic device behaves as a capacitor on short timescales and preserves the memory of two stable states (efficacies) on long timescales. The transitions (LTP/LTD) are stochastic because both the number and the distribution of neural spikes in any finite (stimulation) interval fluctuate, even at fixed pre- and postsynaptic spike rates. The dynamics of the single synapse is studied analytically by extending the solution to a classic problem in queuing theory (Takacs process). The model of the synapse is implemented in aVLSI and consists of only 18 transistors. It is also directly simulated. The simulations indicate that LTP/LTD probabilities versus rates are robust to fluctuations of the electronic parameters in a wide range of rates. The solutions for these probabilities are in very good agreement with both the simulations and measurements. Moreover, the probabilities are readily manipulable by variations of the chip's parameters, even in ranges where they are very small. The tests of the electronic device cover the range from spontaneous activity (3-4 Hz) to stimulus-driven rates (50 Hz). Low transition probabilities can be maintained in all ranges, even though the intrinsic time constants of the device are short (approximately 100 ms). Synaptic transitions are triggered by elevated presynaptic rates: for low presynaptic rates, there are essentially no transitions. The synaptic device can preserve its memory for years in the absence of stimulation. Stochasticity of learning is a result of the variability of interspike intervals; noise is a feature of the distributed dynamics of the network. The fact that the synapse is binary on long timescales solves the stability problem of synaptic efficacies in the absence of stimulation. Yet stochastic learning theory ensures that it does not affect the collective behavior of the network, if the transition probabilities are low and LTP is balanced against LTD.

---

109. Modeling Alternation to Synchrony with Inhibitory Coupling: A Neuromorphic VLSI Approach

Gennady Cymbalyuk, Girish N. Patel, Ronald L. Calabrese, Stephen P. DeWeerth, Avis H. Cohen

https://doi.org/10.1162/089976600300014926 | pp. 2259-2278

We developed an analog very large-scale integrated system of two mutually inhibitory silicon neurons that display several different stable oscillations. For example, oscillations can be synchronous with weak inhibitory coupling and alternating with relatively strong inhibitory coupling. All oscillations observed experimentally were predicted by bifurcation analysis of a corresponding mathematical model. The synchronous oscillations do not require special synaptic properties and are apparently robust enough to survive the variability and constraints inherent in this physical system. In biological experiments with oscillatory neuronal networks, blockade of inhibitory synaptic coupling can sometimes lead to synchronous oscillations. An example of this phenomenon is the transition from alternating to synchronous bursting in the swimming central pattern generator of lamprey when synaptic inhibition is blocked by strychnine. Our results suggest a simple explanation for the observed oscillatory transitions in the lamprey central pattern generator network: that inhibitory connectivity alone is sufficient to produce the observed transition.

---

110. Representation of Concept Lattices by Bidirectional Associative Memories

Radim B■lohlávek

https://doi.org/10.1162/089976600300014935 | pp. 2279-2290

This article presents a concept interpretation of patterns for bidirectional associative memory (BAM) and a representation of hierarchical structures of concepts (concept lattices) by BAMs. The constructive representation theorem provides a storing rule for a training set that allows a concept interpretation. Examples demonstrating the theorems are presented.

---

111. A Silicon Implementation of the Fly's Optomotor Control System

Reid R. Harrison, Christof Koch

https://doi.org/10.1162/089976600300014944 | pp. 2291-2304

Flies are capable of stabilizing their body during free flight by using visual motion information to estimate self-rotation. We have built a hardware model of this optomotor control system in a standard CMOS VLSI process. The result is a small, low-power chip that receives input directly from the real world through on-board photoreceptors and generates motor commands in real time. The chip was tested under closed-loop conditions typically used for insect studies. The silicon system exhibited stable control sufficiently analogous to the biological system to allow for quantitative comparisons.

---

112. Efficient Event-Driven Simulation of Large Networks of Spiking Neurons and Dynamical Synapses

Maurizio Mattia, Paolo Del Giudice

https://doi.org/10.1162/089976600300014953 | pp. 2305-2329

A simulation procedure is described for making feasible large-scale simulations of recurrent neural networks of spiking neurons and plastic synapses. The procedure is applicable if the dynamic variables of both neurons and synapses evolve deterministically between any two successive spikes. Spikes introduce jumps in these variables, and since spike trains are typically noisy, spikes introduce stochasticity into both dynamics. Since all events in the simulation are guided by the arrival of spikes, at neurons or synapses, we name this procedure event-driven. The procedure is described in detail, and its logic and performance are compared with conventional (synchronous) simulations. The main impact of the new approach is a drastic reduction of the computational load incurred upon introduction of dynamic synaptic efficacies, which vary organically as a function of the activities of the pre- and postsynaptic neurons. In fact, the computational load per neuron in the presence of the synaptic dynamics grows linearly with the number of neurons and is only about 6% more than the load with fixed synapses. Even the latter is handled quite efficiently by the algorithm. We illustrate the operation of the algorithm in a specific case with integrate-and-fire neurons and specific spike-driven synaptic dynamics. Both dynamical elements have been found to be naturally implementable in VLSI. This network is simulated to show the effects on the synaptic structure of the presentation of stimuli, as well as the stability of the generated matrix to the neural activity it induces.

---

113. Clustering Irregular Shapes Using High-Order Neurons

Hod Lipson, Hava T. Siegelmann

https://doi.org/10.1162/089976600300014962 | pp. 2331-2353

This article introduces a method for clustering irregularly shaped data arrangements using high-order neurons. Complex analytical shapes are modeled by replacing the classic synaptic weight of the neuron by high-order tensors in homogeneous coordinates. In the first- and second-order cases, this neuron corresponds to a classic neuron and to an ellipsoidalmetric neuron. We show how high-order shapes can be formulated to follow the maximum-correlation activation principle and permit simple local Hebbian learning. We also demonstrate decomposition of spatial arrangements of data clusters, including very close and partially overlapping clusters, which are difficult to distinguish using classic neurons. Superior results are obtained for the Iris data.

---

114. Learning Chaotic Attractors by Neural Networks

Rembrandt Bakker, J.C. Schouten, C. Lee Giles, Floris Takens, C.M. van den Bleek

https://doi.org/10.1162/089976600300014971 | pp. 2355-2383

An algorithm is introduced that trains a neural network to identify chaotic dynamics from a single measured time series. During training, the algorithm learns to short-term predict the time series. At the same time a criterion, developed by Diks, van Zwet, Takens, and de Goede (1996) is monitored that tests the hypothesis that the reconstructed attractors of model-generated and measured data are the same. Training is stopped when the prediction error is low and the model passes this test. Two other features of the algorithm are (1) the way the state of the system, consisting of delays from the time series, has its dimension reduced by weighted principal component analysis data reduction, and (2) the user-adjustable prediction horizon obtained by "error propagation"-partially propagating prediction errors to the next time step. The algorithm is first applied to data from an experimental-driven chaotic pendulum, of which two of the three state variables are known. This is a comprehensive example that shows how well the Diks test can distinguish between slightly different attractors. Second, the algorithm is applied to the same problem, but now one of the two known state variables is ignored. Finally, we present a model for the laser data from the Santa Fe time-series competition (set A). It is the first model for these data that is not only useful for short-term predictions but also generates time series with similar chaotic characteristics as the measured data.

---

115. Generalized Discriminant Analysis Using a Kernel Approach

G. Baudat, F. Anouar

https://doi.org/10.1162/089976600300014980 | pp. 2385-2404

We present a new method that we call generalized discriminant analysis (GDA) to deal with nonlinear discriminant analysis using kernel function operator. The underlying theory is close to the support vector machines (SVM) insofar as the GDA method provides a mapping of the input vectors into high-dimensional feature space. In the transformed space, linear properties make it easy to extend and generalize the classical linear discriminant analysis (LDA) to nonlinear discriminant analysis. The formulation is expressed as an eigenvalue problem resolution. Using a different kernel, one can cover a wide class of nonlinearities. For both simulated data and alternate kernels, we give classification results, as well as the shape of the

decision function. The results are confirmed using real data to perform seed classification.

---

116. Generalization and Selection of Examples in Feedforward Neural Networks

Leonardo Franco, Sergio A. Cannas

https://doi.org/10.1162/089976600300014999 | pp. 2405-2426

In this work, we study how the selection of examples affects the learning procedure in a boolean neural network and its relationship with the complexity of the function under study and its architecture. We analyze the generalization capacity for different target functions with particular architectures through an analytical calculation of the minimum number of examples needed to obtain full generalization (i.e., zero generalization error). The analysis of the training sets associated with such parameter leads us to propose a general architecture-independent criterion for selection of training examples. The criterion was checked through numerical simulations for various particular target functions with particular architectures, as well as for random target functions in a nonoverlapping receptive field perceptron. In all cases, the selection sampling criterion lead to an improvement in the generalization capacity compared with a pure random sampling. We also show that for the parity problem, one of the most used problems for testing learning algorithms, only the use of the whole set of examples ensures global learning in a depth two architecture. We show that this difficulty can be overcome by considering a tree-structured network of depth 2log2(N)-1.

---

117. Stationary and Integrated Autoregressive Neural Network Processes

Adrian Trapletti, Friedrich Leisch, Kurt Hornik

https://doi.org/10.1162/089976600300015006 | pp. 2427-2450

We consider autoregressive neural network (AR-NN) processes driven by additive noise and demonstrate that the characteristic roots of the shortcuts-the standard conditions from linear time-series analysis-determine the stochastic behavior of the overall AR-NN process. If all the characteristic roots are outside the unit circle, then the process is ergodic and stationary. If at least one characteristic root lies inside the unit circle, then the process is transient. AR-NN processes with characteristic roots lying on the unit circle exhibit either ergodic, random walk, or transient behavior. We also analyze the class of integrated AR-NN (ARI-NN) processes and show that a standardized ARI-NN process "converges" to a Wiener process. Finally, least-squares estimation (training) of the stationary models and testing for nonstationarity is discussed. The estimators are shown to be consistent, and expressions on the limiting distributions are given.

---

118. Learning to Forget: Continual Prediction with LSTM

Felix A. Gers, Jürgen Schmidhuber, Fred Cummins

https://doi.org/10.1162/089976600300015015 | pp. 2451-2471

Long short-term memory (LSTM; Hochreiter & Schmidhuber, 1997) can solve numerous tasks not solvable by previous learning algorithms for recurrent neural networks (RNNs). We identify a weakness of LSTM networks processing continual input streams that are not a priori segmented into subsequences with

explicitly marked ends at which the network's internal state could be reset. Without resets, the state may grow indefinitely and eventually cause the network to break down. Our remedy is a novel, adaptive "forget gate" that enables an LSTM cell to learn to reset itself at appropriate times, thus releasing internal resources. We review illustrative benchmark problems on which standard LSTM outperforms other RNN algorithms. All algorithms (including LSTM) fail to solve continual versions of these problems. LSTM with forget gates, however, easily solves them, and in an elegant way.

---

119. Dynamical Mechanism for Sharp Orientation Tuning in an Integrate-and-Fire Model of a Cortical Hypercolumn

Paul C. Bressloff, Neil W. Bressloff, Jack D. Cowan

https://doi.org/10.1162/089976600300014809 | pp. 2473-2511

Orientation tuning in a ring of pulse-coupled integrate-and-fire (IF) neurons is analyzed in terms of spontaneous pattern formation. It is shown how the ring bifurcates from a synchronous state to a non-phase-locked state whose spike trains are characterized by clustered but irregular fluctuations of the interspike intervals (ISIs). The separation of these clusters in phase space results in a localized peak of activity as measured by the time-averaged firing rate of the neurons. This generates a sharp orientation tuning curve that can lock to a slowly rotating, weakly tuned external stimulus. Under certain conditions, the peak can slowly rotate even to a fixed external stimulus. The ring also exhibits hysteresis due to the subcritical nature of the bifurcation to sharp orientation tuning. Such behavior is shown to be consistent with a corresponding analog version of the IF model in the limit of slow synaptic interactions. For fast synapses, the deterministic fluctuations of the ISIs associated with the tuning curve can support a coefficient of variation of order unity.

---

120. Statistical Signs of Common Inhibitory Feedback with Delay

Quentin Pauluis

https://doi.org/10.1162/089976600300014818 | pp. 2513-2518

Cross-correlation histograms (CCHs) sometimes exhibit an isolated central peak flanked by two troughs. What can cause this pattern? The absence of CCH satellite peak makes an oscillatory common input doubtful. It is here shown using a simple counting model that a common inhibitory feedback with delay can account for this pattern.

---

121. On the Computational Power of Winner-Take-All

Wolfgang Maass

https://doi.org/10.1162/089976600300014827 | pp. 2519-2535

This article initiates a rigorous theoretical analysis of the computational power of circuits that employ modules for computing winner-take-all. Computational models that involve competitive stages have so far been neglected in computational complexity theory, although they are widely used in computational brain models, artificial neural networks, and analog VLSI. Our theoretical analysis shows that winner-take-all is a surprisingly powerful computational module in comparison with threshold gates (also referred to as

McCulloch-Pitts neurons) and sigmoidal gates. We prove an optimal quadratic lower bound for computing winner-take-all in any feedforward circuit consisting of threshold gates. In addition we show that arbitrary continuous functions can be approximated by circuits employing a single soft winner-take-all gate as their only nonlinear operation. Our theoretical analysis also provides answers to two basic questions raised by neurophysiologists in view of the well-known asymmetry between excitatory and inhibitory connections in cortical circuits: how much computational power of neural networks is lost if only positive weights are employed in weighted sums and how much adaptive capability is lost if only the positive weights are subject to plasticity.

---

122. Reclassification as Supervised Clustering

A. Sierra, Fernando Corbacho

https://doi.org/10.1162/089976600300014836 | pp. 2537-2546

In some branches of science, such as molecular biology, classes may be defined but not completely trusted. Sometimes posterior analysis proves them to be partially incorrect. Despite its relevance, this phenomenon has not received much attention within the neural computation community. We define reclassification as the task of redefining some given classes by maximum likelihood learning in a model that contains both supervised and unsupervised information. This approach leads to supervised clustering with an additional complexity penalizing term on the number of new classes. As a proof of concept, a simple reclassification algorithm is designed and applied to a data set of gene sequences. To test the performance of the algorithm, two of the original classes are merged. The algorithm is capable of unraveling the original three-class hidden structure, in contrast to the unsupervised version (K-means); moreover, it predicts the subdivision of one of the original classes into two different ones.

---

123. A Model of Invariant Object Recognition in the Visual System: Learning Rules, Activation Functions, Lateral Inhibition, and Information-Based Performance Measures

Edmund T. Rolls, T. Milward

https://doi.org/10.1162/089976600300014845 | pp. 2547-2572

VisNet2 is a model to investigate some aspects of invariant visual object recognition in the primate visual system. It is a four-layer feedforward network with convergence to each part of a layer from a small region of the preceding layer, with competition between the neurons within a layer and with a trace learning rule to help it learn transform invariance. The trace rule is a modified Hebbian rule, which modifies synaptic weights according to both the current firing rates and the firing rates to recently seen stimuli. This enables neurons to learn to respond similarly to the gradually transforming inputs it receives, which over the short term are likely to be about the same object, given the statistics of normal visual inputs. First, we introduce for VisNet2 both single-neuron and multiple-neuron information-theoretic measures of its ability to respond to transformed stimuli. Second, using these measures, we show that quantitatively resetting the trace between stimuli is not necessary for good performance. Third, it is shown that the sigmoid activation functions used in VisNet2, which allow the sparseness of the representation to be controlled, allow good performance when using sparse distributed representations. Fourth, it is shown that VisNet2 operates well with medium-range lateral inhibition with a radius in the same order of size as the region of the preceding layer from which neurons

receive inputs. Fifth, in an investigation of different learning rules for learning transform invariance, it is shown that VisNet2 operates better with a trace rule that incorporates in the trace only activity from the preceding presentations of a given stimulus, with no contribution to the trace from the current presentation, and that this is related to temporal difference learning.

---

124. An Analysis of Orientation and Ocular Dominance Patterns in the Visual Cortex of Cats and Ferrets

Thomas Müller, M. Stetter, Mark Hübener, Frank Sengpiel, Tobias Bonhoeffer, Imke Gödecke, B. Chapman, Siegrid Löwel, Klaus Obermayer

https://doi.org/10.1162/089976600300014854 | pp. 2573-2595

We report an analysis of orientation and ocular dominance maps that were recorded optically from area 17 of cats and ferrets. Similar to a recent study performed in primates (Obermayer & Blasdel, 1997), we find that 80% (for cats and ferrets) of orientation singularities that are nearest neighbors have opposite sign and that the spatial distribution of singularities deviates from a random distribution of points, because the average distances between nearest neighbors are significantly larger than expected for a random distribution. Orientation maps of normally raised cats and ferrets show approximately the same typical wavelength; however, the density of singularities is higher in ferrets than in cats. Also, we find the well-known overrepresentation of cardinal versus oblique orientations in young ferrets (Chapman & Bonhoeffer, 1998; Coppola, White, Fitzpatrick, & Purves, 1998) but only a weak, not quite significant overrepresentation of cardinal orientations in cats, as has been reported previously (Bonhoeffer & Grinvald, 1993). Orientation and ocular dominance slabs in cats exhibit a tendency of being orthogonal to each other (Hubener, Shoham, Grinvald, & Bonhoeffer, 1997), albeit less pronounced, as has been reported for primates (Obermayer & Blasdel, 1993). In chronic recordings from single animals, a decrease of the singularity density and an increase of the ocular dominance wavelength with age but no change of the orientation wavelengths were found. Orientation maps are compared with two pattern models for orientation preference maps: bandpass-filtered white noise and the field analogy model. Bandpass-filtered white noise predicts sign correlations between orientation singularities, but the correlations are significantly stronger (87% opposite sign pairs) than what we have found in the data. Also, bandpass-filtered noise predicts a deviation of the spatial distribution of singularities from a random dot pattern. The field analogy model can account for the structure of certain local patches but not for the whole orientation map. Differences between the predictions of the field analogy model and experimental data are smaller than what has been reported for primates (Obermayer & Blasdel, 1997), which can be explained by the smaller size of the imaged areas in cats and ferrets.

---

125. Improvements to the Sensitivity of Gravitational Clustering for Multiple Neuron Recordings

Stuart N. Baker, George L. Gerstein

https://doi.org/10.1162/089976600300014863 | pp. 2597-2620

We outline two improvements to the technique of gravitational clustering for detection of neuronal synchrony, which are capable of improving the method's detection of weak synchrony with limited data. The advantages of the enhancements are illustrated using data with known levels of synchrony and different

interspike interval distributions. The novel simulation method described can easily generate such test data. An important dependence of the sensitivity of gravitational clustering to the interspike interval distribution of the analysed spike trains is described.

---

126. Neural Coding: Higher-Order Temporal Patterns in the Neurostatistics of Cell Assemblies

Laura Martignon, Gustavo Deco, Kathryn Blackmond Laskey, Mathew E. Diamond, Winrich A. Freiwald, Eilon Vaadia

https://doi.org/10.1162/089976600300014872 | pp. 2621-2653

Recent advances in the technology of multiunit recordings make it possible to test Hebb's hypothesis that neurons do not function in isolation but are organized in assemblies. This has created the need for statistical approaches to detecting the presence of spatiotemporal patterns of more than two neurons in neuron spike train data. We mention three possible measures for the presence of higher-order patterns of neural activation--coefficients of log-linear models, connected cumulants, and redundancies--and present arguments in favor of the coefficients of log-linear models. We present test statistics for detecting the presence of higher-order interactions in spike train data by parameterizing these interactions in terms of coefficients of log-linear models. We also present a Bayesian approach for inferring the existence or absence of interactions and estimating their strength. The two methods, the frequentist and the Bayesian one, are shown to be consistent in the sense that interactions that are detected by either method also tend to be detected by the other. A heuristic for the analysis of temporal patterns is also proposed. Finally, a Bayesian test is presented that establishes stochastic differences between recorded segments of data. The methods are applied to experimental data and synthetic data drawn from our statistical models. Our experimental data are drawn from multiunit recordings in the prefrontal cortex of behaving monkeys, the somatosensory cortex of anesthetized rats, and multiunit recordings in the visual cortex of behaving monkeys.

---

127. Gaussian Processes for Classification: Mean-Field Algorithms

Manfred Opper, Ole Winther

https://doi.org/10.1162/089976600300014881 | pp. 2655-2684

We derive a mean-field algorithm for binary classification with gaussian processes that is based on the TAP approach originally proposed in statistical physics of disordered systems. The theory also yields an approximate leave-one-out estimator for the generalization error, which is computed with no extra computational cost. We show that from the TAP approach, it is possible to derive both a simpler "naive" mean-field theory and support vector machines (SVMs) as limiting cases. For both mean-field algorithms and support vector machines, simulation results for three small benchmark data sets are presented. They show that one may get state-of-the-art performance by using the leave-one-out estimator for model selection and the built-in leave-one-out estimators are extremely precise when compared to the exact leave-one-out estimate. The second result is taken as strong support for the internal consistency of the mean-field approach.

---

128. The Bayesian Evidence Scheme for Regularizing Probability-Density Estimating Neural Networks

Dirk Husmeier

https://doi.org/10.1162/089976600300014890 | pp. 2685-2717

Training probability-density estimating neural networks with the expectation-maximization (EM) algorithm aims to maximize the likelihood of the training set and therefore leads to overfitting for sparse data. In this article, a regularization method for mixture models with generalized linear kernel centers is proposed, which adopts the Bayesian evidence approach and optimizes the hyperparameters of the prior by type II maximum likelihood. This includes a marginalization over the parameters, which is done by Laplace approximation and requires the derivation of the Hessian of the log-likelihood function. The incorporation of this approach into the standard training scheme leads to a modified form of the EM algorithm, which includes a regularization term and adapts the hyperparameters on-line after each EM cycle. The article presents applications of this scheme to classification problems, the prediction of stochastic time series, and latent space models.

---

129. A Bayesian Committee Machine

Volker Tresp

https://doi.org/10.1162/089976600300014908 | pp. 2719-2741

The Bayesian committee machine (BCM) is a novel approach to combining estimators that were trained on different data sets. Although the BCM can be applied to the combination of any kind of estimators, the main foci are gaussian process regression and related systems such as regularization networks and smoothing splines for which the degrees of freedom increase with the number of training data. Somewhat surprisingly, we find that the performance of the BCM improves if several test points are queried at the same time and is optimal if the number of test points is at least as large as the degrees of freedom of the estimator. The BCM also provides a new solution for on-line learning with potential applications to data mining. We apply the BCM to systems with fixed basis functions and discuss its relationship to gaussian process regression. Finally, we show how the ideas behind the BCM can be applied in a non-Bayesian setting to extend the input-dependent combination of estimators.

---

130. Evolution of Cooperative Problem Solving in an Artificial Economy

Eric B. Baum, Igor ■ur■anovi■

https://doi.org/10.1162/089976600300014700 | pp. 2743-2775

We address the problem of how to reinforce learning in ultracomplex environments, with huge state-spaces, where one must learn to exploit a compact structure of the problem domain. The approach we propose is to simulate the evolution of an artificial economy of computer programs. The economy is constructed based on two simple principles so as to assign credit to the individual programs for collaborating on problem solutions. We find empirically that starting from programs that are random computer code, we can develop systems that solve hard problems. In particular, our economy learned to solve almost all random Blocks World problems with goal stacks that are 200 blocks high. Competing methods solve such problems only up to goal stacks of at most 8 blocks. Our economy has also learned to unscramble about half a randomly scrambled Rubik's cube and to solve several commercially sold puzzles.

131. Cortical Potential Distributions and Information Processing

Henry C. Tuckwell

https://doi.org/10.1162/089976600300014719 | pp. 2777-2795

The use of sets of spatiotemporal cortical potential distributions (CPDs) as the basis for cognitive information processing results in a very large space of cognitive elements with natural metrics. Results obtained from current source density (CSD) analysis suggest that in the CPD picture, action potentials may make only a relatively minor contribution to the brain's code. In order to establish if two CPDs are close, we consider standard metrics in spaces of continuous functions, and these may be employed to ascertain if two stimuli will be identified as the same. The correspondence between the electrical activity within brain regions, including not only action potentials but all postsynaptic potentials (PSPs), and CPDs is considered. We examine the possibility of using the CSD approach to find potential distributions using the descriptive approach in which precise sets of times are ascribed to the occurrence of action potentials and PSPs. Using metrics in the multidimensional space of paths of collections of point processes, we show that closeness of CPDs is implied by closeness of sets of spike times and PSP times if a certain metric is used but not others. We also set forth a dynamical model consisting of a system of reaction-dif fusion equations for ionic concentrations coupled with nerve membrane potential equations and active transport systems. Making the approximation of a descriptive approach, the correspondence between sets of spike times and PSP times and CPDs is obtained as with the CSD method. However, since it is not possible to ascribe precise times to the occurrence of PSPs and action potentials, the descriptive approach cannot be used to describe the configuration of electrical activity in cortical regions accurately. We also discuss how the CPD framework relates to the binding problem and submillisecond timing.

132. Asymptotic Bias in Information Estimates and the Exponential (Bell) Polynomials

Jonathan D. Victor

https://doi.org/10.1162/089976600300014728 | pp. 2797-2804

We present a new derivation of the asymptotic correction for bias in the estimate of information from a finite sample. The new derivation reveals a relationship between information estimates and a sequence of polynomials with combinatorial significance, the exponential (Bell) polynomials, and helps to provide an understanding of the form and behavior of the asymptotic correction for bias.

133. Relating Macroscopic Measures of Brain Activity to Fast, Dynamic Neuronal Interactions

D. Chawla, Erik D. Lumer, Karl Friston

https://doi.org/10.1162/089976600300014737 | pp. 2805-2821

In this article we used biologically plausible simulations of coupled neuronal populations to address the relationship between phasic and fast coherent neuronal interactions and macroscopic measures of activity that are integrated over time, such as the BOLD response in functional magnetic resonance imaging. Event-related, dynamic correlations were assessed using joint peristimulus time histograms and, in

particular, the mutual information between stimulus-induced transients in two populations. This mutual information can be considered as an index of functional connectivity. Our simulations showed that functional connectivity or dynamic integration between two populations increases with mean background activity and stimulus-related rate modulation. Furthermore, as the background activity increases, the populations become increasingly sensitive to the intensity of the stimulus in terms of a predisposition to transient phase locking. This reflects an interaction between background activity and stimulus intensity in producing dynamic correlations, in that background activity augments stimulus-induced coherence modulation. This is interesting from a computational perspective because background activity establishes a context that may have a profound effect on event-related interactions or functional connectivity between neuronal populations. Finally, total firing rates, which subsume both background activity and stimulus-related rate modulation, were almost linearly related to the expression of dynamic correlations over large ranges of activities. These observations show that under the assumptions implicit in our model, rate-specific metrics based on rate or coherence modulation may be different perspectives on the same underlying dynamics. This suggests that activity (averaged over all peristimulus times), as measured in neuroimaging, may be tightly coupled to the expression of dynamic correlations.

---

134. Analysis of Pointing Errors Reveals Properties of Data Representations and Coordinate Transformations Within the Central Nervous System

Joseph McIntyre, F. Stratta, Jacques Droulez, Francesco Lacquaniti

https://doi.org/10.1162/089976600300014746 | pp. 2823-2855

The execution of a simple pointing task invokes a chain of processing that includes visual acquisition of the target, coordination of multimodal proprioceptive signals, and ultimately the generation of a motor command that will drive the finger to the desired target location. These processes in the sensorimotor chain can be described in terms of internal representations of the target or limb positions and coordinate transformations between different internal reference frames. In this article we first describe how different types of error analysis can be used to identify properties of the internal representations and coordinate transformations within the central nervous system. We then describe a series of experiments in which subjects pointed to remembered 3D visual targets under two lighting conditions (dim light and total darkness) and after two different memory delays (0.5 and 5.0 s) and report results in terms of variable error, constant error, and local distortion. Finally, we present a set of simulations to help explain the patterns of errors produced in this pointing task. These analyses and experiments provide insight into the structure of the underlying sensorimotor processes employed by the central nervous system.

---

135. Modeling Selective Attention Using a Neuromorphic Analog VLSI Device

Giacomo Indiveri

https://doi.org/10.1162/089976600300014755 | pp. 2857-2880

Attentional mechanisms are required to overcome the problem of flooding a limited processing capacity system with information. They are present in biological sensory systems and can be a useful engineering tool for artificial visual systems. In this article we present a hardware model of a selective attention mechanism implemented on a very large-scale integration (VLSI) chip, using analog neuromorphic circuits. The chip exploits a spike-based representation to receive, process, and transmit signals. It can be used as a

transceiver module for building multichip neuromorphic vision systems. We describe the circuits that carry out the main processing stages of the selective attention mechanism and provide experimental data for each circuit. We demonstrate the expected behavior of the model at the system level by stimulating the chip with both artificially generated control signals and signals obtained from a saliency map, computed from an image containing several salient features.

---

136. Asymptotic Convergence Rate of the EM Algorithm for Gaussian Mixtures

Jinwen Ma, Lei Xu, Michael I. Jordan

https://doi.org/10.1162/089976600300014764 | pp. 2881-2907

It is well known that the convergence rate of the expectation-maximization (EM) algorithm can be faster than those of convention first-order iterative algorithms when the overlap in the given mixture is small. But this argument has not been mathematically proved yet. This article studies this problem asymptotically in the setting of gaussian mixtures under the theoretical framework of Xu and Jordan (1996). It has been proved that the asymptotic convergence rate of the EM algorithm for gaussian mixtures locally around the true solution Theta* is o(e(0. 5-epsilon)(Theta*)), where epsilon > 0 is an arbitrarily small number, o(x) means that it is a higher-order infinitesimal as x --> 0, and e(Theta*) is a measure of the average overlap of gaussians in the mixture. In other words, the large sample local convergence rate for the EM algorithm tends to be asymptotically superlinear when e(Theta*) tends to zero.

---

137. Incremental Active Learning for Optimal Generalization

Masashi Sugiyama, Hidemitsu Ogawa

https://doi.org/10.1162/089976600300014773 | pp. 2909-2940

The problem of designing input signals for optimal generalization is called active learning. In this article, we give a two-stage sampling scheme for reducing both the bias and variance, and based on this scheme, we propose two active learning methods. One is the multipoint search method applicable to arbitrary models. The effectiveness of this method is shown through computer simulations. The other is the optimal sampling method in trigonometric polynomial models. This method precisely specifies the optimal sampling locations.

---

138. A Quantitative Study of Fault Tolerance, Noise Immunity, and Generalization Ability of MLPs

J. L. Bernier, Julio Ortega, Eduardo Ros, Ignacio Rojas, A. Prieto

https://doi.org/10.1162/089976600300014782 | pp. 2941-2964

An analysis of the influence of weight and input perturbations in a multilayer perceptron (MLP) is made in this article. Quantitative measurements of fault tolerance, noise immunity, and generalization ability are provided. From the expressions obtained, it is possible to justify some previously reported conjectures and experimentally obtained results (e.g., the influence of weight magnitudes, the relation between training with noise and the generalization ability, the relation between fault tolerance and the generalization ability). The measurements introduced here are explicitly related to the mean squared error degradation in the presence

of perturbations, thus constituting a selection criterion between different alternatives of weight configurations. Moreover, they allow us to predict the degradation of the learning performance of an MLP when its weights or inputs are deviated from their nominal values and thus, the behavior of a physical implementation can be evaluated before the weights are mapped on it according to its accuracy.

---

139. On the Computational Complexity of Binary and Analog Symmetric Hopfield Nets

Ji■■■ Š■■ma, Pekka Orponen, Teemu AnttiPoika

https://doi.org/10.1162/089976600300014791 | pp. 2965-2989

We investigate the computational properties of finite binary- and analog-state discrete-time symmetric Hopfield nets. For binary networks, we obtain a simulation of convergent asymmetric networks by symmetric networks with only a linear increase in network size and computation time. Then we analyze the convergence time of Hopfield nets in terms of the length of their bit representations. Here we construct an analog symmetric network whose convergence time exceeds the convergence time of any binary Hopfield net with the same representation length. Further, we prove that the MIN ENERGY problem for analog Hopfield nets is NP-hard and provide a polynomial time approximation algorithm for this problem in the case of binary nets. Finally, we show that symmetric analog nets with an external clock are computationally Turing universal.

---

140. Index

(authors unknown)

https://doi.org/10.1162/neco.2000.12.12.2991 | pp. 2991-3010

No abstract available.